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ABSTRACT

Higher trade policy uncertainty has recessionary effects on U.S. states. To demonstrate this, we first build a novel empirical measure of regional trade policy uncertainty based on the volatility of national import tariffs at the sectoral level and on the sectoral composition of imports in U.S. states. We find that a state that is more exposed to an unanticipated increase in tariff volatility suffers from a larger drop in real GDP and employment than the average U.S. state. We then build a two-region open-economy model and find that the precautionary saving behavior is the main driver of the recession, although this effect is reinforced by high exposure to import tariffs. The feedback effect resulting from trade connections with the Foreign country primarily influences the persistence of these dynamics.

1. Introduction

In recent years, many countries have experienced economic and political disturbances initiated by the revision of trade agreements and a general fostering of trade protectionism. The most prominent events were the Brexit vote in 2016 and the trade war launched by the Trump Administration in 2018. More recently, the additional import tariffs pledged by Donald Trump in his second presidency are expected to fuel economic volatility. These events led to an exceptional boost in uncertainty regarding future market conditions, especially in the import–export market. The impact of higher trade policy uncertainty, which might result

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from unpredictable trade agreement negotiations or rising volatility in import tariffs, has been extensively analyzed in the literature (see [Handley and Limão \(2022\)](#) for a survey). Trade policy uncertainty has individual effects on firms' investment, employment, and export decisions, as well as aggregate effects on GDP at the country level. While tariffs on imports are decided at the national level, one might suspect that changes in tariffs and the uncertainty surrounding them have different effects at the regional level, depending on the composition of imports in each region. In this study, we address this question by analyzing the effects of uncertainty on trade policy at the regional level, focusing on the U.S. state level.

One contribution of this paper is to offer a novel empirical measure of trade policy uncertainty at the U.S. state level. We capture trade policy uncertainty using the volatility of import tariffs. First, we construct a national series of tariffs at the sectoral level based on the share of duties (collected by U.S. customs) over imports for each sector. We then estimate the time-varying volatility of these sectoral tariff series using a stochastic volatility model, as in [Born and Pfeifer \(2014\)](#) and [Fernández-Villaverde et al. \(2015\)](#) for fiscal policy. Second, we build exposure to uncertainty at the state level by weighting sector-based volatility by the share of sectoral imports in the total imports within the state. We find for instance that the sectors "Stone and Glass" or "Textile" suffer the most from tariff volatility, and some states, like New-York, are especially exposed to trade policy uncertainty since they intensively import those products. We investigate the effects of uncertainty shocks on U.S. states' economic activity using our U.S. state-level measure of exposure to trade policy uncertainty. Our estimation shows a significant and sizable negative effect of higher exposure to trade policy uncertainty on economic activity, whether measured by employment or GDP growth at the regional level.

We then develop an open two-region model to investigate the mechanisms by which trade policy uncertainty affects the activity of U.S. states. In our model, both regions make up the Home country (the U.S.), and each region trades within the country and with the rest of the world. Both regions tax their imports through a common tariff set nationally but differ in their exposure to international trade. We investigate the dynamics of a representative trade-exposed U.S. state that features a high import share (i.e., the import-to-GDP ratio) and a high degree of export intensity, which we define as the value of exports relative to the total value of exports in the country. The model is calibrated using the empirical evidence emphasized in the empirical section. We simulate the model using trade policy uncertainty shocks to extract a model-based series of output. This strategy allows us to confront the model with the data. We find that our stylized model, when submitted to the same data-generating process as the empirical data, can reproduce our main empirical finding: higher exposure to trade policy uncertainty deteriorates economic activity. The fact that our model is regional allows us to further analyze how exposure to trade affects the transmission channels of shocks to trade policy uncertainty.

We find that the exposed region experiences a recession when hit by volatility shocks to national import tariffs. This result can be explained by the two main drivers that are highlighted in this paper. First, it is well accepted that uncertainty shocks affect the economy through the precautionary behavior of agents. More precisely, because firms in the region are strongly exposed to import tariffs, they are more uncertain about their future marginal costs and thus adopt a precautionary pricing behavior, whereas households are more willing to adopt precautionary saving behavior. As a result, aggregate demand in the Home country drops when trade policy uncertainty increases. Owing to the presence of nominal rigidities in the Home country, prices do not adjust as quickly as they should, leading to a strong recession and a large increase in markups. Second, in the context of an open economy, a fall in aggregate demand has international consequences because it reduces the demand for foreign goods, which affects the region, especially when it is highly exposed to trade. Indeed, as the Foreign country imports less, the exposed region produces less, generating an international trade feedback effect. We perform a counterfactual analysis to better understand these international trade channels. Unsurprisingly, a lower degree of trade openness makes the demand function for imported goods less sensitive to tariff volatility, thus dampening recessions. Interestingly, we find that the international feedback effect, which we can shut down by setting a large home bias in the Foreign country, mostly affects the persistence of the dynamics, notably by causing a long-lasting drop in output in response to the negative trade policy uncertainty shock.

Contributions to the literature

A large body of the literature measures trade policy uncertainty based on various data sources and concepts of uncertainty. [Handley and Limão \(2022\)](#) provide an extensive survey of the tools to measure trade policy uncertainty and its economic effects. For instance, uncertainty can be measured through changes in trade agreements ([Handley, 2014](#); [Handley and Limão, 2015, 2017](#)); textual analysis methods in newspapers, country reports, or earnings conference calls by firms [Baker et al. \(2016\)](#), [Caldara et al. \(2020\)](#), [Ahir et al. \(2022\)](#) or tariff risks ([Alessandria et al., 2024](#)). Our contribution to the literature is to provide a measure of uncertainty at the regional level based on the volatility of import tariffs. This is common practice in macroeconomics using tax volatility as a measure of uncertainty in the tradition of [Born and Pfeifer \(2014\)](#) and [Fernández-Villaverde et al. \(2015\)](#). These authors resort to a stochastic volatility model to estimate the time-varying volatility of taxes. [Caldara et al. \(2020\)](#) apply this process to measure tariff uncertainty at the national level for the U.S. economy. In this study, we examine whether this methodology can be enriched to deliver a measure of tariff uncertainty at the U.S. state level, which we refer to as regional trade policy uncertainty. Thus, we join an important branch of the literature devoted to the disaggregation of uncertainty below the national level. [Baker et al. \(2022\)](#) provide historical series of economic policy uncertainty at the U.S. State level.¹ We differ from them in methodology, since they resort to a textual analysis of local newspapers, which allowed them to determine whether uncertainty comes from local events or national/international events. In addition, we contribute to the literature by providing a U.S. state-level measure of uncertainty specific to trade policies.

¹ [Elkamhi et al. \(2024\)](#) provide a similar index, using a different set of newspapers and removing the nationwide information. [Shoag and Veuger \(2016\)](#) adopt a similar approach focusing on the Great Recession.

While the literature has investigated the aggregate effects of trade policy uncertainty on the economy (Handley and Limão, 2017; Steinberg, 2019; Caldara et al., 2020; Firooz et al., 2025), to the best of our knowledge, it is silent regarding its effects from a regional perspective. There is increasing interest in macroeconomics for regional/state data to understand this global phenomenon – see Chodorow-Reich (2020) for a discussion on the use of regional data in macroeconomics. The interest in regional data is to improve the identification of causal relationships compared with macroeconomic data that generally aggregate several simultaneous events. The literature on regional local fiscal multipliers was reviewed in Chodorow-Reich (2019), including notable contributions by Chodorow-Reich et al. (2012), Nakamura and Steinsson (2014) and Dupor and Guerrero (2017). Theoretical models with a special focus on regional economies have been developed in Beraja et al. (2018, 2019), House et al. (2018) and House et al. (2021). The local effects of trade policy have also been extensively studied in this literature to better understand the consequences of China's integration into international trade (Autor et al., 2013) and the protectionist measures established by the U.S. in 2018 (Fajgelbaum et al., 2020). However, the literature on the local effects of trade policy is scant when it comes to the uncertainty surrounding this policy, partly because of a lack of data. Because we develop a regional measure of uncertainty, we can supplement the literature by considering the role of uncertainty shocks using regional data in a structural open economy model.

The remainder of this paper is organized as follows. In Section 2, we build an empirical measure of regional trade policy uncertainty, and investigate the effects on the U.S. state economies. Section 3 describes the theoretical model, and Section 4 explains its parametrization. Section 5 analyzes the transmission channels of trade policy uncertainty shocks, and Section 6 concludes.

2. Empirical analysis

This section outlines the methodology used to build a novel measure of trade policy uncertainty (TPU, henceforth) based on the sector-specific volatility of import tariffs and highlight the effects of TPU shocks on economic activity at the U.S. state level. We proceed in three steps. First, we construct a measure of tariff uncertainty by extracting the time-varying volatility of (national) imports tariff series at the sectoral level. Second, we use these series to measure exposure to the tariff uncertainty of U.S. states, considering the sectoral composition of imports by state. Third, we resort to a panel estimation to quantify the effects of TPU on U.S. states' economic indicators. All data sources are detailed in the online appendix.

2.1. A measure of tariff volatility

We construct sector-specific series of tariff rate by using data from the Census Bureau Foreign Trade Database (USA Trade Online). We extract the monthly data on U.S. imports by commodity type.² Using the Harmonized System Codes of the World Customs Organization from 2017, we convert commodity types into 15 sectors.³ The data sample extends from 2008m2 to 2020m12. Let $p = 1, \dots, P$, denote the sector index with $P = 15$ and $t = 1, \dots, T$, the time index. Let $M_{p,t}$ denote the total real value of imports in sector p at time t , and $CD_{p,t}$ are the estimates of the calculated duty. The values of imports and customs duties at the sectoral level were computed by summing all the types of commodity holdings in a sector. All series are seasonally adjusted. We define the gross rate of the national effective tariff at the sectoral level $\tau_{p,t}$ as

$$\tau_{p,t} = 1 + \frac{CD_{p,t}}{M_{p,t}}. \quad (1)$$

Fig. 1 shows the gross import tariff rate $\tau_{p,t}$ for the 15 sample sectors. Unsurprisingly, most sectors experienced a surge in their import tariffs during the U.S. Trade War, initiated in 2018 by President D. Trump, mostly against China but also against other countries. The first wave of tariffs increase started in Spring in 2018, when the Federal government started imposing tariffs on steel and aluminum imports against several countries, including China and Canada, which led to an average tariff rate in the metal sector of 1.2% in February 2018 and 7.8% in March 2019. In September 2019, supplementary tariffs were imposed on imports targeting textile products from China, leading to an average tariff rate in the textile sector of 11.9% in August 2019 and 14.9% in September 2019. Despite this exceptional period of high volatility, Fig. 1 also shows tariff heterogeneity across sectors for the full sample. For example, in September 2009, the Obama Administration signed China-specific safeguard actions against tire imports over three years by imposing tariffs of 35% in the first year, 30% in the second, and 25% in the third year.⁴ Fig. 1 reports this protective trade policy through the average tariff rate in the "Plastic and Rubbers" increased from 2.1% in August 2009 to 2.8% in December 2009.

The stochastic volatility model is an attractive setup for disentangling level shocks from volatility shocks and therefore extract time-varying uncertainty (see Fernández-Villaverde and Guerrón-Quintana (2020) for discussion). In addition, this statistical process

² Note that the database covers trade in manufacturing and the agricultural sectors, excluding trade in services. We use imports for consumption, i.e. "Measures the total of merchandise that has been physically cleared through Customs either entering consumption channels immediately or entering after withdrawal for consumption from bonded warehouses or Foreign Trade Zones under U.S. Customs and Border Protection (CBP) custody" (source: Census).

³ The 15 sectors are classified using the 2-digit HS code list: "01-05 Animal & Animal Products," "06-15 Vegetable Products," "16-24 Foodstuffs," "25-27 Mineral Products," "28-38 Chemicals & Allied Industries," "39-40 Plastics/Rubbers," "41-43 Raw Hides, Skins, Leather, & Furs," "44-49 Wood & Wood Products," "50-63 Textiles," "64-67 Footwear/Headgear," "68-71 Stone/Glass," "72-83 Metals," "84-85 Machinery/Electrical," "86-89 Transportation," "90-97 Miscellaneous." The Harmonized System Codes of the World Customs Organization of 2017 are natural candidates because our raw data are classified using a Harmonized System. Santacreu et al. (2023) also construct a sectoral classification, in which they aggregate their raw data using NAICS (2012) classification. We share more than 70% of their 16 sectors.

⁴ These tariffs were applied to products 4011.10.10, 4011.10.50, 4011.20.10, and 4011.20.50 of the Harmonized Tariff Schedule of the United States. See Bown and Kolb (2023) for a discussion of the reform.

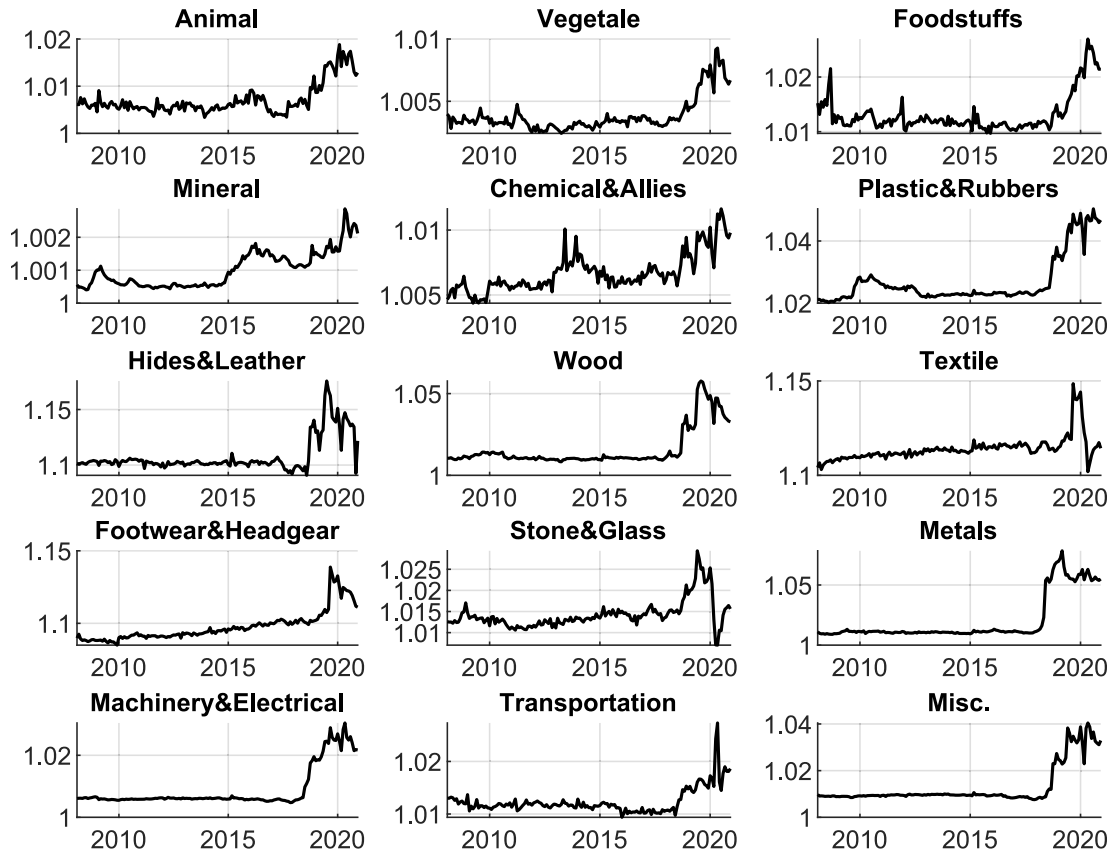


Fig. 1. Sector-specific gross tariff rate. Note: The sector-specific gross tariff rate, $\tau_{p,t}$, is defined in Eq. (1).

is in line with the typical way of modeling time-varying volatility shocks in business cycle type of models (see Section 3). Fernández-Villaverde et al. (2011) apply this methodology to extract time-varying volatility in real interest rates (Fernández-Villaverde et al., 2015) on fiscal instruments and like us, Caldara et al. (2020) estimate a stochastic volatility process for import tariffs. In this spirit, we assume that $\Delta\tau_{p,t}$, the monthly growth rate of $\tau_{p,t}$, follows an AR(1) process with time-varying volatility

$$\Delta\tau_{p,t} = \rho_p^\tau \Delta\tau_{p,t-1} + \alpha_p \Delta\tau_{-p,t} + \exp(\sigma_{p,t}^\tau) \varepsilon_{p,t}^\tau, \quad (2)$$

with

$$\sigma_{p,t}^\tau = (1 - \rho_p^\sigma) \bar{\sigma}_p^\tau + \rho_p^\sigma \sigma_{p,t-1}^\tau + \eta_p \varepsilon_{p,t}^\sigma, \quad (3)$$

where $\Delta\tau_{p,t}$ is demeaned, $\varepsilon_{p,t}^\tau \sim \mathcal{N}(0, 1)$ and $\varepsilon_{p,t}^\sigma \sim \mathcal{N}(0, 1)$ for $p = 1, \dots, P$. Let $\sigma_{p,t}^\tau$ denote the measure of trade policy uncertainty in sector p . Following Fernández-Villaverde et al. (2011), $\sigma_{p,t}^\tau$ is expressed as exponential in Eq. (2) to ensure that trade policy uncertainty remains positive. Level (first-order) shocks, $\varepsilon_{p,t}^\tau$, correspond to unexpected variations in the sector- p tariff growth rate and volatility (second-order) shocks, $\varepsilon_{p,t}^\sigma$, correspond to unexpected variations in its standard deviation. The parameter ρ_p^τ drives the persistence of level shocks, and ρ_p^σ drives persistence of volatility shocks. $\bar{\sigma}_p^\tau$ determines the average standard deviation of innovations $\varepsilon_{p,t}^\tau$ and η_p governs the magnitude of tariff volatility shocks. The process (2)–(3) is estimated sector-by-sector. As we are interested in sector-specific volatility, we control for common variations in tariffs from other sectors than p . Specifically, we add the gross effective tax rates for all sectors but p and compute the associated variation denoted by $\Delta\tau_{-p,t}$. Thus, parameter α_p captures variations in $\Delta\tau_{p,t}$ which are explained by the aggregate dynamics of all other sectors.

Regressions (2)–(3) capture dynamics of sectoral growth rate tariffs as well as their time-varying standard deviation. Caldara et al. (2020) estimate a similar stochastic volatility process on the U.S. aggregate series of tariffs, whereas we consider the sector-specific tariff uncertainty. By using the algorithm proposed in Born and Pfeifer (2014), Eqs. (2)–(3) are estimated jointly with Bayesian methods using sequential Monte Carlo Methods. We obtain priors directly from the data using simple least-squares regressions. The historical series of the unobserved volatility shocks are extracted using a nonlinear particle filter from a posterior distribution. The posterior distributions of the parameters are computed from 20 000 draws where the first 5 000 are discarded. The Metropolis–Hastings parameter is adjusted to get an acceptance rate of approximately 25%. The posterior estimates are reported in the online appendix.

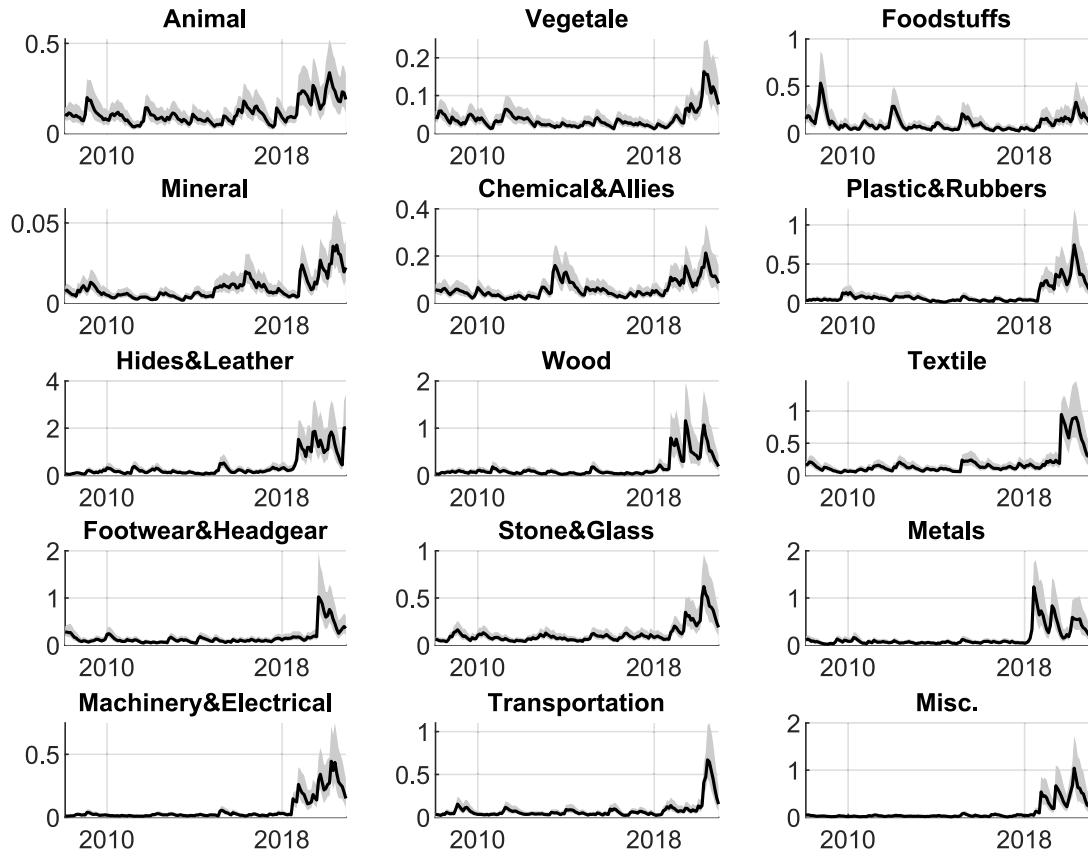


Fig. 2. Sector-specific tariff volatility series.

Note: The tariff volatility $\sigma_{p,t}^{\tau}$ is defined in Eq. (3). The displayed series are $(100 \times \exp(\sigma_{p,t}^{\tau}))$ based on the posterior estimation results reported in the online appendix. The solid line is the median historical smoothed series and the grey area displays the [10th, 90th] percentile probability interval.

Fig. 2 shows the historical series of tariff uncertainty as estimated in Eq. (3). The solid line represents the median historical smoothed estimate of $\sigma_{p,t}^{\tau}$, expressed in exponential and multiplied by 100. The grey area displays the first and last deciles, respectively. The figure confirms that if tariff volatility is sector specific, all sectors experienced a peak in volatility from 2018. As mentioned above, “Footwear and Headgear”, “Foodstuffs”, and “Textiles” display high volatility on average. However, some sectors exhibit a particularly high volatility in tariffs during the 2018 Trade War peak, such as the “Stone and Glass” or “Metals”. Subsequently, we exploit this heterogeneity across sectors to capture U.S. states’ exposure to trade policy uncertainty.⁵

2.2. State-level exposure to uncertainty

In the spirit of the shift-share design, we assess the effects of trade policy uncertainty in U.S. states using the weighted average of sectoral shocks as regressor, with the weight being the state-exposure share (see Adão et al. (2019) and Goldsmith-Pinkham et al. (2020) for the recent discussion on shift-share settings). This strategy allows us to proxy state-level TPU shocks by assigning volatility shocks to sectoral import composition. In practice, we build a series of exposure to TPU shocks in state- s ($TPU_{s,t}$), weighting volatility shocks ($\varepsilon_{p,t}^{\sigma}$) by the state-specific sectoral composition of imports ($\mu_{p,s,t}$) as

$$\mu_{p,s,t} \equiv \frac{M_{p,s,t}}{M_{s,t}}. \quad (4)$$

$M_{p,s,t}$ corresponds to sector- p custom values collected from imports in state s at time t and $\mu_{p,s,t}$ is the share of sector- p imports in the total imports for state s at time t . Series of imports at the state level, $M_{p,s,t}$, are extracted from the Census Bureau Foreign Trade

⁵ In the online appendix, we compare each sector-specific tariff volatility series with the newspaper-based index of TPU for the U.S. provided by Caldara et al. (2020). On average, the correlation coefficient is 0.33, and it varies between -0.03 and 0.68 depending on the sector. These correlation coefficients are significantly different from zero for most sectors (except for the foodstuffs sector). Each peak in the news-based TPU index is associated with an increase in stochastic volatilities specific to certain sectors. Therefore, our sectoral series on TPU allows us to decompose much of the information contained in this aggregate TPU indicator according to the sectors concerned.

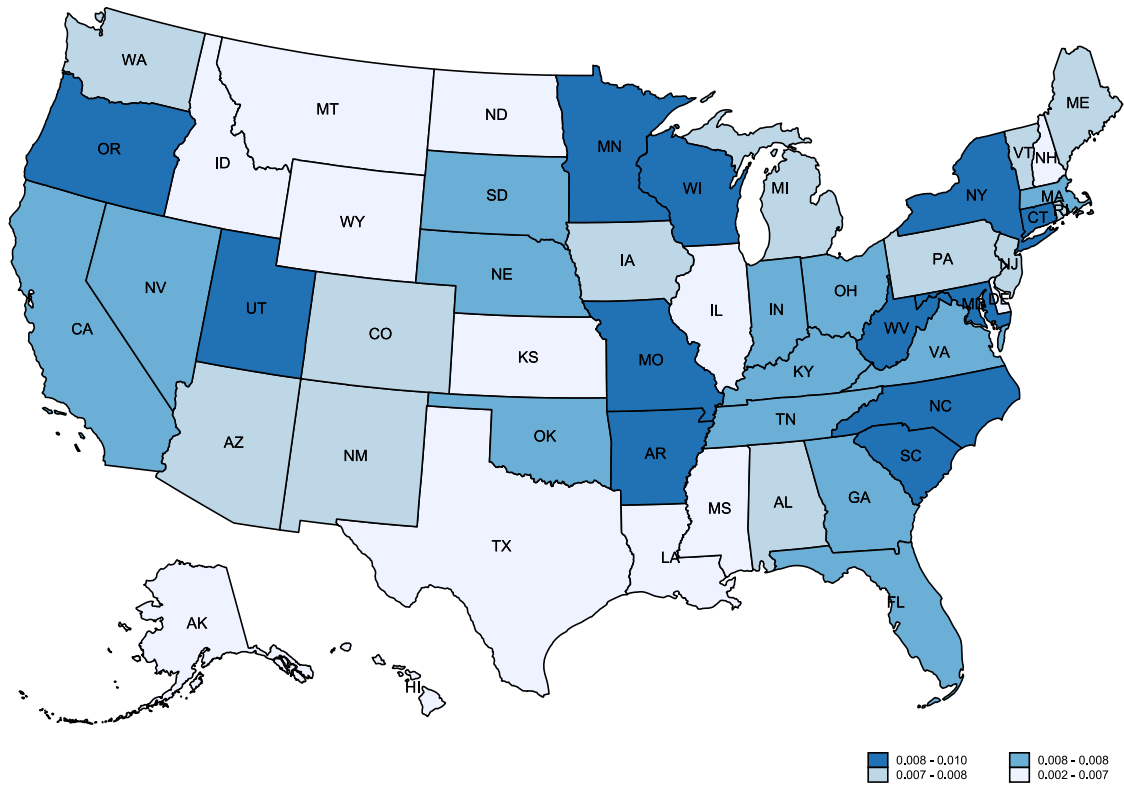


Fig. 3. Tariff-volatility exposure across U.S. states.

Note: Colors illustrate states' exposure to tariff volatility, which are computed plugging σ_p^e in Eq. (5) and averaging over the sample 2008m2-2020m12. Darker colors represent states which are the most exposed to tariff volatility.

Database and available at a monthly frequency. In addition, we do not only capture the sectoral composition of imports but also the state's degree of openness by using the state's overall import share in the weight. Therefore, we define $\omega_{s,t} \equiv M_{s,t}/Y_{s,t}$ as the state-level share of imports in the GDP computed at a yearly frequency. Finally, to mitigate endogeneity issues, we use the constant share ($\bar{\mu}_{p,s} \times \bar{\omega}_s$) over the first three years of the sample (2009–2011) for each state. The series of exposure to TPU shocks in states ($TPU_{s,t}$) is

$$TPU_{s,t} = \sum_{p=1}^P (\bar{\mu}_{p,s} \bar{\omega}_s) \times \varepsilon_{p,t}^{\sigma} \quad (5)$$

where $s = 1, \dots, 50$, $\bar{\mu}_{p,s} = (1/36) \times \sum_{t=1}^{36} \mu_{p,s,t}$ using the definition of $\mu_{p,s,t}$ given by Eq. (4), $\bar{\omega}_s = (1/3) \sum_{t=1}^3 \omega_{s,t}$ and $\varepsilon_{p,t}^{\sigma}$ is estimated by using Eq. (3).

Fig. 3 plots the average degree of exposure to tariff volatility for each state. Darker colors correspond to the states that are most exposed to sectors with high volatility in import tariffs. The most exposed state is New York (NY) which is also among the top five largest importers in the U.S. Note that 28% of imports reaching NY comes from the “Stone and Glass” and 11% from the sector “Textile”. As mentioned previously, these two sectors feature high volatilities (see Fig. 2). They also explain the high degree of exposure of Utah (UT)—39% of its imports are “Stone and Glass”—and North Carolina (NC) because 16% of its imports are “Textile”. Sector “Metal” also explains the high exposure to tariff volatility. For example, Connecticut (CT) and Maryland (MD) are the largest importers of this sector (16% and 15% of imports, respectively). As shown in Fig. 2, “Metal” was extensively affected by the 2018 Trade War as its volatility of tariffs jumped.

2.3. Regional effects of trade policy uncertainty

Armed with a state-level measure of exposure to tariff volatility, we investigate the economic effects of TPU shocks.

2.3.1. Baseline estimation

We estimate a cross-sectional regression in the spirit of “local multipliers” literature (see Chodorow-Reich (2019) and Canova (2024) for instance), which allows us to use the heterogeneity across states to assess the dynamic effects of a policy change. We

Table 1
Effects of TPU shocks on the economic activity.

	Real GDP			Employment		
	(I)	(II)	(III)	(IV)	(V)	(VI)
	$h = 0$	$h = 4$	$h = 8$	$h = 0$	$h = 4$	$h = 8$
Volatility exposure (TPU)	−0.042 (0.059)	−0.322* (0.173)	−0.548*** (0.203)	−0.022 (0.051)	−0.277*** (0.092)	−0.292** (0.110)
Level exposure	0.030 (0.024)	0.074 (0.045)	0.074 (0.084)	0.015 (0.014)	0.018 (0.043)	0.025 (0.069)
Δ CI	0.151*** (0.032)	0.896*** (0.211)	0.792** (0.331)	0.334*** (0.042)	0.666*** (0.169)	0.739*** (0.252)
Observations	2,550	2,350	2,150	2,550	2,350	2,150
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: Table reports the effects of TPU shocks on the real GDP in Columns (I)–(III) and on employment in Columns (IV)–(VI). Each column reports the results from regression (6) for the horizon $h = 0, 4, 8$ quarters. The coefficient of interest β_h associated to $TPU_{s,t}$ is reported in the first line. All regressions include state- and time-FE. Robust standard errors are clustered at the state level.

** $p < 0.01$. * $p < 0.05$. * $p < 0.1$.

estimate the following baseline regression

$$y_{s,t+h} - y_{s,t-1} = \delta_{s,h} + \delta_{t,h} + \beta_h TPU_{s,t} + \gamma_h X_{s,t} + v_{s,t+h}, \quad (6)$$

where β_h corresponds to the cumulative response of the outcome variable $y_{s,t+h}$ $h \geq 0$ periods after the realization of the TPU exposure shock in state s .⁶ We consider two measures of economic activity $y_{s,t}$, namely real GDP and employment. The former consists of state-level GDP in private industries and is available from the Bureau of Economic Analysis. Employment is the total number of employees in the private sector as reported by the Bureau of Labor Statistics. Both measures are expressed in log, and Eq. (6) is estimated weighting observations using the population share of states in 2008, as in Dupor and Guerrero (2017). Because the real state-level GDP is available at a quarterly frequency, we convert our series of volatility exposures $TPU_{s,t}$, from monthly to quarterly, using an averaging transformation. We include time fixed effects, $\delta_{t,h}$, to absorb all common shocks in the U.S., and state fixed effects, $\delta_{s,h}$, to control for unobservable state-specific factors that might affect the outcome variable. To mitigate the potential omitted variable bias, we also include $X_{s,t}$, a vector of state-level control variable. As the first control variable, we include the exposure to level shocks, which corresponds to the state exposure to sectoral tariff shocks, as built in Eq. (5), where we replace $\varepsilon_{p,t}^\sigma$ by $\varepsilon_{p,t}^\tau$. In doing so, we seek to isolate the pure effects of an increase in volatility exposure by removing all variations that would result from a level shock, even though the stochastic process (2)–(3) disentangles the volatility and level shocks, which are supposed to be uncorrelated. Second, we control for the business cycle by using the growth rate of the Coincident Economic Activity Index provided by Federal Reserve Bank of Philadelphia (Δ CI). In the online appendix, we report robustness results with alternative outcome and control variables. The sample ranges from 2008q1 to 2020q4. Eq. (6) is estimated for horizons $h = 0, 4, 8$ by pooling the 50 states over the 64 quarters.⁷

Table 1 presents the estimation results. The first three columns show the relative effects of an exogenous increase in the exposure to TPU on the state-level real GDP at impact ($h = 0$) after a year ($h = 4$) and after two years ($h = 8$), all other horizons are reported in the online appendix. We find that higher exposure to trade policy uncertainty has recessionary effects, although it takes one year to materialize. The maximum impact is achieved after two years with the estimate $\beta_h = -0.548$, which is significant at the 1% level. Considering a high shock of 0.017 (corresponding to the 90th percentile of the shock distribution), this peak corresponds to a 0.93% cumulated drop in output over the two years. When it comes to employment, we also find that a higher exposure to uncertainty significantly reduces employment after a year. The estimates indicate that a shock to an exposure of 0.017 reduces employment by 0.49% after eight quarters and it is highly significant.

Thus, trade policy uncertainty generates persistent and large magnitude effects on the economic activity of U.S. states. The magnitudes of these effects are assessed by controlling for the state's cyclical situation at the time of the shock as measured by the change in the coincident index Δ CI, which is naturally highly correlated with the future evolution of the output and employment in states. The last interesting result from Table 1 is that higher exposure to tariffs (in level) does not significantly affect economic activity for the considered horizons h . We find a positive and significant impact on output, in quarter-1 only (see online appendix).

⁶ Notice that β_h cannot be interpreted as the aggregate response at the U.S. level. It measures the relative change in the outcome variable between the two states in response to an increase in TPU exposure in a treated (or shocked) region relative to an untreated region. Therefore, by resorting to state-level estimation, we depart from Caldara et al. (2020) who estimate a bivariate SVAR model using U.S. data and showed that positive shocks to tariff volatility are recessionary at the national level.

⁷ Canova (2024) highlights potential bias in cross-sectional estimation of average dynamic effect when data are pooled. Subsequently, we run a state-by-state estimation of Eq. (6), where the coefficient of TPU becomes state-specific and is denoted by $\beta_{s,h}$. As shown in the online appendix, the relatively low dispersion between the estimated coefficients suggests that our baseline estimates are not affected by the bias associated with the heterogeneity between states' responses to trade policy uncertainty.

Table 2
Robustness checks.

	(Ia)	(Ib)	(II)	(III)	(IV)	(V)	(VI)
Volatility exposure (TPU)	−0.548** (0.205)	−0.543** (0.208)	−0.264* (0.142)	−0.523** (0.231)	−0.103*** (0.025)	−0.428* (0.237)	—
Level exposure	0.077 (0.084)	0.064 (0.086)	−0.004 (0.102)	0.073 (0.084)	0.009 (0.015)	−0.007 (0.083)	0.038 (0.094)
Δ CI	0.788** (0.329)	0.775** (0.328)	0.797*** (0.257)	0.791** (0.333)	0.802** (0.330)	0.836** (0.336)	0.799** (0.332)
Geographical spillovers	−0.008 (0.007)						
Trade spillovers		0.032*** (0.011)					
Retaliation effect			−2.819 (2.737)				
Swing-state interaction				−0.092 (0.267)			
TPU placebo							−0.004 (0.008)
Obs.	2,150	2,150	1,200	2,150	2,150	1,950	2,150
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Each column reports the results from regression (6) for the horizon $h = 8$ quarters on real GDP. The coefficient of interest β_h associated to $TPU_{s,t}$ is reported in the first line. Columns (Ia)–(Ib)–(II)–(III) extend the benchmark regression reported in Column (III) of Table 1 by sequentially adding the variables associated to geographic spillovers, trade spillovers, retaliation effect, and swing-state interaction (see text for definitions and sources). Column (IV) reports the results when the exposure to uncertainty is weighted using the intermediate goods intensity to import. Column (V) corresponds to the pre-2017 sample estimation. Column (VI) is the placebo regression of the benchmark regression. All regressions include state- and time-FE. Robust standard errors are clustered at the state level.

** $p < 0.01$. * $p < 0.05$. * $p < 0.1$.

Notice that the stochastic volatility process is supposed to disentangle the source of variations in import tariffs between changes in levels and changes in volatilities. It is clear from Eq. (2) that a rise in tariffs can be explained by a rise in $\epsilon_{p,t}^r$, a specific realization of the innovation in levels drawn from a distribution with constant variance and/or by a rise in $\sigma_{p,t}^r$, that magnifies the impact of a given realization of $\epsilon_{p,t}^r$. Our estimates suggest that a large part of the changes in tariff is attributed to volatility while level shocks play a less important role. We report in the online appendix the results when we re-estimate Eq. (2) with $\sigma_{p,t}^r$ being constant. The process becomes a standard AR(1) process, implying that we attribute *all* variations of tariffs to levels shocks. We show that the point estimates are larger (although still positive). This finding confirms that our estimation based on the stochastic volatility model identifies a sequence of volatility shocks whose economic effects differ from those of level shocks. This leads us to conclude that an increase in import tariffs might have small (and positive) effects on regional economies.

2.3.2. Robustness checks

We now explore the robustness of our findings. Table 2 reports the estimation results for several alternative specifications. For brevity, we focus on the effects of TPU shocks on real GDP after two years ($h = 8$). In the online appendix, we report the estimation results for all horizons and on both GDP and employment.

(I). Spillover Effects

The cross-sectional dimension of our analysis helps us to assess the extent to which stronger exposure of a state to TPU shocks alters its economic activity relative to the average state's exposure. Therefore, it fails to capture the spillover effects across units –U.S. states here– that might emerge because these units are interconnected (see Auerbach et al. (2020), for instance). We check the robustness of our results when we consider the geographical and trade spillover effects across states. One state can be affected not only by its own idiosyncratic TPU shocks, but also indirectly by TPU shocks hitting neighboring states. In this case, our estimation does not fully estimate the economic impact of TPU shocks because these indirect effects are not considered. First, we enrich the baseline regression (6) by adding a new variable that corresponds to the sum of all TPU exposure indices ($TPU_{s,t}$) of the neighboring states. Column (Ia) of Table 2 shows that TPU shocks in neighboring states do not significantly affect the economic activity of a specific state. This finding suggests that geographical spillovers are small. Interestingly, when we assume that TPU shocks do not come from neighboring states, but from the main economic partner of one state, as defined by Dupor and Guerrero (2017), we find a strong positive effect (see Column (Ib) of Table 2). However, our main estimates are not altered and remain significant. This positive spillover effect suggests that the total effect of trade policy uncertainty is smaller than that of our benchmark.

(II). Retaliation Effects

Any changes in the level and volatility of import tariffs may lead trading partners to retaliate. In order to control for the reaction of foreign countries, we re-run the benchmark regression by including the increase in retaliation tariff as provided by Fajgelbaum et al. (2020), which is sector specific. More precisely, we aggregate their measure of the retaliation tax increase over the 15 sectors presented in Section 2.1 and build a state-specific retaliation exposure index using the state's export share to

total U.S. exports. Due to data availability, we re-run the estimation over a shorter sample (2013q1–2019q2). Column (II) in Table 2 shows that the state's GDP reacts negatively, but not significantly, to the retaliation tax in quarter 8. More importantly, stronger exposure to TPU still significantly reduces GDP, whereas the less pronounced effect is mostly explained by the shorter sample size.

(III). Swing-states Effects

The last alternative control variable that we consider relates to political considerations. To check the exogeneity of the tariff measures volatility exposure, we investigate the role of swing states in our baseline regression. We use data from the MIT Election Data and Science Lab (MEDSL) which provides state-level returns for elections to the U.S. presidency from 1976 to 2020. We create a dummy equal to one when a state is considered as a swing state; that is, the election returns have shifted from Democrats to Republicans (or vice-versa) at least once over the period 2008–2020. The swing states include Arizona (AZ), Florida (FL), Georgia (GA), Indiana (IN), Iowa (IA), Michigan (MI), North Carolina (NC), Ohio (OH), Pennsylvania (PA), and Wisconsin (WI). We then create the interaction term between the tariff volatility exposure series and the swing-states dummy. This interaction term is negative and significantly different from zero at impact (reported in the online appendix), but not at longer horizons, as shown in Column (III) of Table 2. The point estimates of β_h is close to that reported in our benchmark regression.

(IV). Intermediate Goods Share

The series of exposure to TPU shocks built from Eq. (5) is based on the sectoral composition of total imports at the state level, abstracting from whether goods are final or intermediate. We investigate whether our results hold when we considered the share of intermediate goods in imports using the Input-Output database provided by WIOT in 2008.⁸ We then compute the share of intermediate goods imported in the U.S., relative to the total number of imported goods, and weigh the state-based exposure to uncertainty by the intermediate goods intensity to imports by state. Column (IV) in Table 2 shows that the effect of volatility exposure on GDP remains significant. Note that the magnitude cannot be directly compared because the scales of the shocks differ between this specification and our benchmark.

(V). Sample Without Trade War

We estimate the baseline regression over the sample 2008q1–2017q4 to abstract from the 2018 Trade War. As shown in Column (V) of Table 2, the estimated coefficients are slightly lower and significantly different from zero only at the 10% level for real GDP against 5% for the benchmark, but remained significant at the 1% level for employment (reported in the online appendix). It is remarkable that even if we discard the trade war episodes from the sample, we still find significant economic effects of TPU shocks. This suggests that the recessionary effects of trade policy uncertainty is not only driven by the exceptional jump in tariff volatility that occurred in 2018–2020.

(VI). Placebo Shocks

The degree of exposure to trade policy uncertainty is based on the state-specific sectoral composition of imports, as explained in Eq. (4). To check whether our results capture exogenous variations in tariff volatility instead of being driven by the heterogeneous composition of imports, we run a placebo test by randomly resampling the vector of sectoral tariff volatility, and building a counterfeit measure of exposure to TPU. Column (VI) in Table 2 presents the results. We find that when the baseline regression is estimated using this placebo series, volatility exposure shocks no longer have a significant effect on states' economic activity.

2.3.3. Effects of TPU shocks on inflation

Our previous findings showed that higher exposure to trade policy uncertainty significantly reduces state-level economic activity. However, a full picture of the effects of state-based TPU requires an assessment of how prices react to higher uncertainty because stronger volatility on import taxes is likely to alter expected firms' marginal costs and, therefore, price dynamics. To conduct this analysis, we use the state-specific quarterly inflation rates for the Consumer Price Index (CPI) provided by Hazell et al. (2022). Owing to limited data availability, the sample covers the period 2008q1–2017q4. However, we showed in Section 2.3.2 that our results are not entirely driven by the 2018 Trade War. In addition, CPI inflation series are provided for 32 states, which altogether represent 86% of total GDP in the U.S.

Table 3 presents the results when we estimate regression (6) by setting the price index as the outcome variable, $y_{s,t}$, built from the CPI inflation measure provided by Hazell et al. (2022).

The effect of the tariff level shock on trade policy is positive and close to significance at 10%. This positive sign is in line with the transmission of tariff shocks to prices e.g., Fajgelbaum et al. (2020). Regarding the effects of uncertainty shocks on trade policy, we find that cumulative inflation becomes significantly negative after two years. This positive co-movement between output and prices suggests that import tariff uncertainty shocks look like demand shocks, as argued in Leduc and Liu (2016) and Basu and Bundick (2017). In the next section, we develop a regional open-economy model consistent with the negative effects of TPU shocks on output and inflation.

3. Theoretical model

We now describe the theoretical model which allows us to go further in the understanding of the transmission channels of TPU shocks at the regional level. The economy is made up of two countries, Home and Foreign. The Home country is split into two

⁸ We thank our discussant, Joseph Steinberg, for this suggestion.

Table 3
Effects of TPU shocks on inflation.

	(I) $h = 0$	(II) $h = 4$	(III) $h = 8$
Volatility exposure (TPU)	−0.013 (0.110)	−0.643* (0.337)	−0.916** (0.363)
Level exposure	−0.030 (0.018)	−0.026 (0.077)	−0.032 (0.111)
Δ CI	0.103 (0.070)	−0.242 (0.325)	−1.142*** (0.414)
Observations	1,287	1,155	1,023
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Note: Table reports the effects of TPU shocks on inflation in Columns (I)–(III). Each column reports the results from regression (6) for the horizon $h = 0, 4, 8$ quarters. The coefficient of interest β_h associated to $TPU_{s,t}$ is reported in the first line. All regressions include state- and time-FE. Robust standard errors are clustered at the state level.

** p<0.01. * p<0.05. * p<0.1.

groups of regions of size η_j where $j = \{1, 2\}$ is the region-specific index. The Foreign country is considered as a whole, his size being η^* , such that we normalize $\eta_1 + \eta_2 + \eta^* = 1$. The two Home regions belong to a monetary union. We assume that individual goods are tradable at the interstate and the international level with the Foreign country. For international trade, we assume that the Home country impose national import taxes on imported intermediate goods. We abstract from labor and capital mobility. The online appendix provides details of the computations.

3.1. Production sector

In each region j of the Home country, at time t , a representative firm produces a non-tradable final good ($Y_{j,t}$) by aggregating differentiated intermediate goods. Intermediate goods are produced by monopolistic competitors who combine domestic and imported individual goods through a CES production function. They face price adjustment costs à la Calvo (1983) and we consider a producer currency pricing setting implying that the law of one price holds. Finally, the individual goods are produced in each region by a representative wholesaler who uses labor and capital. In the Foreign country, the production structure is the same, the representative final good producer uses a combination of domestic and imported individual goods, imported from both regions. Foreign variables are denoted with a star *. For notation convenience of trade-related variables, the lower index of these variables corresponds to the country/region of origin (for the production of goods) and the upper index corresponds to the country/region of destination. Also, when we describe the production sector in Region j , we refer to the other region using index j^- . We start describing in details the production in the Home country and then we present briefly the Foreign country when it differs.

3.1.1. Final good production

In Region j of the Home country, a representative firm produces a final good $Y_{j,t}$, by combining type- i differentiated intermediate goods, $Y_{j,t}(i)$ using a Dixit-Stiglitz production function

$$Y_{j,t} = \left[\int_0^1 Y_{j,t}(i)^{\frac{\epsilon_p - 1}{\epsilon_p}} di \right]^{\frac{\epsilon_p}{\epsilon_p - 1}}, \quad (7)$$

where $\epsilon_p > 1$ is the elasticity of substitution between intermediate goods. Let $P_{j,t}$ denote the aggregate price index of the Home Region- j final good $Y_{j,t}$ and $P_{j,t}(i)$ denote the price of type- i region- j intermediate good, $Y_{j,t}(i)$. The profit maximization yields the typical demand function for intermediate goods $Y_{j,t}(i)$

$$Y_{j,t}(i) = \rho_i^j(i)^{-\epsilon_p} Y_{j,t}, \quad (8)$$

where $\rho_{j,t}(i) \equiv P_{j,t}(i) / P_{j,t}$ is the relative price of differentiated goods.

3.1.2. Intermediate good production

In Region j of the Home country, we assume a continuum of monopolistic competitive firms, indexed by $i \in [0, 1]$. Type- i monopolistic competitive firm produces a non-tradable intermediate goods $Y_{j,t}(i)$ by combining domestic and imported individual goods ($\tilde{Y}_{H,t}^j$ and $Y_{F,t}^j$, respectively) through a CES production function

$$Y_{j,t}(i) = \left[(1 - \omega_j)^{\frac{1}{\theta}} (\tilde{Y}_{H,t}^j)^{\frac{\theta - 1}{\theta}} + (\omega_j)^{\frac{1}{\theta}} (Y_{F,t}^j)^{\frac{\theta - 1}{\theta}} \right]^{\frac{\theta}{\theta - 1}}, \quad (9)$$

where $\theta > 1$ is the trade elasticity of substitution between domestic and foreign goods and parameter $\omega_j \in [0, 1]$ is the degree of openness, i.e. it drives the magnitude of the home bias in Region j . Domestic goods correspond to a bundle of goods, $\tilde{Y}_{H,t}^j$, coming from its own location ($Y_{H,t}^j$) and from the other region ($Y_{H,t}^{j-}$), such that

$$\tilde{Y}_{H,t}^j = \left[(\hat{\omega}_j)^{\frac{1}{\theta}} \left(Y_{H,t}^j \right)^{\frac{\theta-1}{\theta}} + (1 - \hat{\omega}_j)^{\frac{1}{\theta}} \left(Y_{H,t}^{j-} \right)^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}, \quad (10)$$

where $\hat{\omega}_j$ measures the degree of home bias of each region relative to the other region.

In a first step, the type- i monopolistic producer from Region j chooses the amount of domestic and imported individual goods, $Y_{H,t}^j$, $Y_{H,t}^{j-}$ and $Y_{F,t}^j$ resp., so as to minimize its real cost, denoted by $S_t^j(i)$, taking into account the production function (9), with

$$S_t^j(i) \equiv \frac{P_{H,t}^j}{P_t^j(i)} Y_{H,t}^j + \frac{P_{H,t}^{j-}}{P_t^{j-}(i)} \frac{P_t^{j-}(i)}{P_t^j(i)} Y_{H,t}^{j-} + (1 + \tau_t) \frac{P_{F,t}}{P_t^j(i)} Y_{F,t}^j, \quad (11)$$

where $P_{H,t}^j$ and $P_{H,t}^{j-}$ is the price index of the good produced in Region j and j^- respectively, and $P_{F,t}$ denote the price index of the individual goods produced by the Foreign country.⁹ Since the tariff policy at Home is national, this rate is not region-specific. The minimization program yields the demand for domestic individual goods, expressed at the symmetric equilibrium,

$$Y_{H,t}^j = (1 - \omega_j) \hat{\omega}_j \left(\frac{\rho_{H,t}^j}{\Lambda_t^j} \right)^{-\theta} Y_t^j, \quad \text{and} \quad Y_{H,t}^{j-} = (1 - \omega_j) (1 - \hat{\omega}_j) \left(\frac{\rho_{H,t}^{j-} Q_{j,t}}{Q_{j,t} \Lambda_t^j} \right)^{-\theta} Y_t^j, \quad (12)$$

as well as the demand for imported individual goods, expressed at the symmetric equilibrium,

$$Y_{F,t}^j = \omega_j \left(\frac{(1 + \tau_t) \rho_{F,t}^j}{\Lambda_t^j} \right)^{-\theta} Y_t^j, \quad (13)$$

where $\Lambda_{j,t}$ denotes the real marginal cost of the intermediate good producer in Region j , $\rho_{H,t}^j \equiv P_{H,t}^j / P_t^j$ and $Q_{j,t}$ is the state-based real exchange rate defined below.

In a second stage, the type- i monopolistic competitor in Region j chooses the price of its good, $P_{j,t}(i)$, by maximizing its profits $d_{j,t}(i)$. We assume that price setting is modeled through a Calvo price setting. Precisely, a type- i intermediate good producer keeps its previous price with probability α_p and resets its price with probability $1 - \alpha_p$. This gives the standard non-linear New-Keynesian Phillips Curve (NKPC)

$$\pi_{j,t}^* = \pi_{j,t} \left(\frac{\varepsilon_p}{\varepsilon_p - 1} \right) \frac{A_{j,t}}{B_{j,t}}, \quad (14)$$

where $\pi_{j,t}^* \equiv P_{j,t}^* / P_{j,t-1}$ and $P_{j,t}^*$ denotes the optimal reset price, all expressed at the symmetric equilibrium. In addition, $A_{j,t}$ and $B_{j,t}$ are defined recursively as

$$A_{j,t} = \Lambda_{j,t} Y_{j,t} + \alpha_p E_t \left\{ \beta_{j,t,t+1} (\pi_{j,t+1})^{\varepsilon_p} A_{j,t+1} \right\}, \quad (15)$$

$$B_{j,t} = Y_{j,t} + \alpha_p E_t \left\{ \beta_{j,t,t+1} (\pi_{j,t+1})^{\varepsilon_p - 1} B_{j,t+1} \right\}, \quad (16)$$

where $\beta_{j,t,t+1}$ is the stochastic discount factor of the household and E_t is the expectation operator conditional to the information available in period t . Expressing the aggregate price index in terms of inflation rate yields

$$\pi_{j,t}^{1-\varepsilon_p} = (1 - \alpha_p) (\pi_{j,t}^*)^{1-\alpha_p} + \alpha_p. \quad (17)$$

Notice that we assume zero inflation at the steady-state. The real marginal cost $\Lambda_{j,t}$ in Region j is increasing with import tariff, τ_t , all things equal, with

$$\Lambda_{j,t} = \rho_{H,t}^j \frac{Y_{H,t}^j}{Y_{j,t}} + \frac{\rho_{H,t}^{j-} Q_{j,t}}{Q_{j,t}} \frac{Y_{H,t}^{j-}}{Y_{j,t}} + (1 + \tau_t) \rho_{F,t}^j \frac{Y_{F,t}^j}{Y_{j,t}}. \quad (18)$$

As standard in the literature, the NKPC given by Eq. (14) tells us that monopolistic competitive firms set their price taking into account the future expected real marginal cost. In our regional open-economy model, a region which is less open (ω_j small) will have its real marginal cost less sensitive to the import tax τ_t (see Eqs. (12) and (18)).

3.1.3. Wholesaler

In Region j of the Home country, we assume perfectly competitive wholesale producers who produce the good $X_{j,t}$ by using capital services, $k_{j,t}$, and labor $\ell_{j,t}$. The wholesale good is then split into intermediate goods used domestically, denoted by $Y_{H,t}^j$, and abroad, denoted by $Y_{H,t}^{j*}$, such that

$$X_{j,t} = Y_{H,t}^j + Y_{H,t}^{j*}. \quad (19)$$

⁹ The imported quantity of good $Y_{F,t}^j$ may differ according to the region j , but the price $P_{F,t}$ is identical for the two regions. Similarly, the price of goods produced in Region j is the same however its location of consumption, $P_{H,t}^j = P_{H,t}^{j-} = P_{H,t}^j$.

The supply of intermediate goods used domestically is equal to the demand by monopolistic producers: $Y_{H,t}^j = \int_0^1 Y_{H,t}^j(i) di$. The wholesale good $X_{j,t}$ is produced with a Cobb–Douglas production function

$$X_{j,t} = a_t \eta_j k_{j,t}^\alpha \ell_{j,t}^{1-\alpha}, \quad (20)$$

where a_t corresponds to a total factor productivity (TFP) shock, defined below. The Region- j size, η_j , intervenes since the input quantities, $k_{j,t}$ and labor $\ell_{j,t}$, are per household of Region j .

The representative wholesaler chooses the amount of capital and labor services, as well as the supply of intermediate goods, so as to maximize its real profits denoted by $\tilde{d}_{j,t}$, subject to the production function (20) and definition (19). The real profits write

$$\tilde{d}_{j,t} = \rho_{Hj,t} Y_{H,t}^j + Q_{j,t} \rho_{Hj,t}^* Y_{Hj,t}^* - w_{j,t} \eta_j \ell_{j,t} - z_{j,t} \eta_j k_{j,t}, \quad (21)$$

where $\rho_{Hj,t}^* \equiv P_{Hj,t}^*/P_t^*$, $w_{j,t}$ and $z_{j,t}$ denote the Region- j real wage and rental rate of capital, respectively. The real exchange rate in Region j is defined as the price of Foreign consumption goods basket relative to the price of Region- j consumption goods basket, such that $Q_{j,t} \equiv e_t P_t^*/P_{j,t}$, where e_t is the nominal exchange rate. The law of one price holds, such that

$$\rho_{Hj,t} = Q_{j,t} \rho_{Hj,t}^*. \quad (22)$$

Therefore, intermediate good producers charge the same price for both their domestic and their import markets. Profit maximization yields the labor-demand and capital-demand equations

$$w_{j,t} = (1 - \alpha) \rho_{Hj,t} \frac{X_{j,t}}{\eta_j \ell_{j,t}}, \quad \text{and} \quad z_{j,t} = \alpha \rho_{Hj,t} \frac{X_{j,t}}{\eta_j k_{j,t}}. \quad (23)$$

3.1.4. Foreign country

We now turn to describe the production sector in the Foreign country. For the sake of simplicity, we keep the Foreign country as stylized as possible, abstracting for instance from nominal rigidities. The Foreign market creates an additional trade connection between the two regions of the Home country since the Foreign final good, Y_t^* , is produced by using both domestic, $Y_{F,t}^*$, and imported individual goods, $\tilde{Y}_{H,t}^*$, using the following production function

$$Y_t^* = \left[(1 - \omega^*)^{\frac{1}{\theta}} (Y_{F,t}^*)^{\frac{\theta-1}{\theta}} + (\omega^*)^{\frac{1}{\theta}} (\tilde{Y}_{H,t}^*)^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}, \quad (24)$$

where $\omega^* \in [0, 1]$ is the Foreign degree of trade openness. We assume that the imported individual good, $\tilde{Y}_{H,t}^*$, is a bundle of imported goods from each Home region

$$\tilde{Y}_{H,t}^* = \left[(\tilde{\omega})^{\frac{1}{\theta}} (Y_{Hj,t}^*)^{\frac{\theta-1}{\theta}} + (1 - \tilde{\omega})^{\frac{1}{\theta}} (Y_{Hj^-,t}^*)^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}. \quad (25)$$

Parameter $\tilde{\omega} \in [0, 1]$ drives the degree of preference of Foreign producers for Region 1's individual goods. Let P_t^* denotes the price of the final good Y_t^* , $P_{F,t}^*$ the price of domestic individual goods $Y_{F,t}^*$, and $P_{Hj,t}^*$ the price of individual goods imported from Region j . The profit maximization of the final good producer gives the demand equation for domestic goods as

$$Y_{F,t}^* = (1 - \omega^*) \left(\rho_{F,t}^* \right)^{-\theta} Y_t^*, \quad (26)$$

with $\rho_{F,t}^* \equiv P_{F,t}^*/P_t^*$, the relative price of domestically produced individual goods, and the demand equations for imported goods as

$$Y_{Hj,t}^* = \omega^* \tilde{\omega} (\rho_{Hj,t}^*)^{-\theta} Y_t^*, \quad \text{and} \quad Y_{Hj^-,t}^* = \omega^* (1 - \tilde{\omega}) (\rho_{Hj^-,t}^*)^{-\theta} Y_t^*, \quad (27)$$

with $\rho_{Hj,t}^* \equiv P_{Hj,t}^*/P_t^*$, the relative price of the imported goods from region j . A stronger preference for Region j ($\tilde{\omega}$) increases the imports of goods coming from this region.

Similarly to the Home country, we assume that a representative wholesaler produces an individual good which is sent domestically ($Y_{F,t}^*$) or exported in each region of the Home country ($Y_{F,t}^1$ and $Y_{F,t}^2$), such that

$$X_t^* = Y_{F,t}^* + Y_{F,t}^1 + Y_{F,t}^2. \quad (28)$$

The good X_t^* is produced using capital, k_t^* , and labor, ℓ_t^* , through the following Cobb–Douglas production function

$$X_t^* = \eta^* (k_t^*)^\alpha (\ell_t^*)^{1-\alpha}. \quad (29)$$

The demand for labor and capital are therefore given by

$$w_t^* = (1 - \alpha) \rho_{F,t}^* \frac{X_t^*}{\eta^* \ell_t^*}, \quad \text{and} \quad z_t^* = \alpha \rho_{F,t}^* \frac{X_t^*}{\eta^* k_t^*}. \quad (30)$$

while the law of one price implies that

$$\rho_{Ft}^* = \frac{\rho_{Ft}^1}{Q_{1,t}}, \quad \text{and} \quad \rho_{Ft}^1 = \frac{Q_{1,t}}{Q_{2,t}} \rho_{Ft}^2. \quad (31)$$

The zero profit condition gives

$$\rho_{Ft}^* Y_{Ft}^* + \frac{\rho_{Ft}^1}{Q_{1,t}} Y_{Ft}^1 + \frac{\rho_{Ft}^2}{Q_{2,t}} Y_{Ft}^2 = w_t^* \ell_t^* + z_t^* k_t^*. \quad (32)$$

3.2. Household

We assume that households make their saving decisions under financial autarky, implying that they do not have access to international lending or borrowing and thus there is no opportunity to share financial risks across borders. For the sake of brevity, we focus here only on the Home country. The Home country is populated by a continuum of identical households. In each region, they choose a sequence of consumption ($c_{j,t}$) and regional bonds ($b_{j,t}$), and they supply labor ($\ell_{j,t}$) and capital services ($k_{j,t}$) to maximize the discounted lifetime utility. The objective of the representative household is thus to maximize

$$E_t \sum_{t=0}^{\infty} \beta^t \left\{ \frac{1}{1-\sigma} \left[\left(c_{j,t} - \psi \ell_{j,t}^{\omega_w} \chi_{j,t} \right)^{1-\sigma} - 1 \right] \right\}, \quad (33)$$

where the term $\chi_{j,t}$ is defined by

$$\chi_{j,t} = c_{j,t}^{\sigma_X} \chi_{j,t-1}^{1-\sigma_X}, \quad (34)$$

with $\beta \in (0, 1)$ is the subjective discount factor, $\sigma_X \in (0, 1)$ drives the strength of the wealth effect on labor supply, ω_w^{-1} is the Frisch elasticity of labor supply, ψ is a normalizing constant (which governs the relative disutility of labor effort) and σ is the inverse of the risk aversion coefficient. We assume that the representative household has access in $t-1$ to a one-period regional bond ($B_{j,t-1}$), which pays off a gross riskless interest rate R_t . She can also save by investing in physical capital by the amount $i_{j,t}$ and she receives labor income, with the real wage denoted $w_{j,t}$, and capital services revenue, with the real return denoted $z_{j,t}$. Last, the household perceives dividend income from differentiated goods producers and lump-sum transfers from the government. Accordingly, the budget constraint writes down as

$$c_{j,t} + b_{j,t} + i_{j,t} = \frac{R_{t-1}}{\pi_{j,t}} b_{j,t-1} + w_{j,t} \ell_{j,t} + z_{j,t} k_{j,t} + T_{j,t} + T_{j,t}^f, \quad (35)$$

where $b_{j,t} = B_{j,t}/P_{j,t}$ denotes the amount of bonds held in real terms, $T_{j,t}$ and $T_{j,t}^f$ are lump-sum transfers of profits from differentiated firms and of the international bond adjustment, respectively. We assume that the law of motion of capital is given by

$$k_{j,t+1} = (1-\delta) k_{j,t} + \left[1 - \frac{\kappa}{2} \left(\frac{i_{j,t}}{i_{j,t-1}} - 1 \right)^2 \right] i_{j,t}, \quad (36)$$

where κ drives the strength of investment adjustment costs. For the sake of space, first-order conditions, which are standard, are reported in [Appendix A](#).

3.3. Fiscal and monetary policy

State-specific government in each country runs a balanced budget

$$\eta_j T_{j,t}^f = \eta_j b_{j,t} - \frac{R_{t-1}}{\pi_{j,t}} \eta_j b_{j,t-1} + \tau \rho_{Ft}^j Y_{Ft}^j - g_t, \quad (37)$$

where g_t corresponds to public spending, which follows an exogenous process as defined below. In addition, all regions in the Home country are part of a monetary union. The Home central bank sets the nominal interest rate rule according to the following Taylor Rule

$$\frac{R_t}{\bar{R}} = \left(\frac{R_{t-1}}{\bar{R}} \right)^{\rho_r} \left(\frac{\pi_t}{\bar{\pi}} \right)^{(1-\rho_r)\rho_\pi} m_t, \quad (38)$$

where ρ_r drives the degree of interest rate inertia and ρ_π is the weight on CPI national inflation, defined as $\pi_t = \frac{\eta_1}{\eta_1+\eta_2} \pi_{1,t} + \frac{\eta_2}{\eta_1+\eta_2} \pi_{2,t}$ and m_t is a monetary policy shock defined below. The real exchange rate at the national level is given by

$$Q_t = (Q_{1,t})^{\frac{\eta_1}{\eta_1+\eta_2}} (Q_{2,t})^{\frac{\eta_2}{\eta_1+\eta_2}}. \quad (39)$$

3.3.1. Equilibrium and shocks

The market clearing equation in Region j on the final goods market

$$Y_{j,t} = \eta_j c_{j,t} + \eta_j i_{j,t} + \eta_j g_t, \quad (40)$$

Table 4
Parametrization.

Common Parameters			
β	Discount factor	0.99	Annual interest rate (4%)
α	Capital share	0.33	Labor income share (0.67)
δ	Capital depreciation rate	0.025	Annual depreciation rate (10%)
ε_p	Elasticity of substitution btw intermediate goods	11	Markup (10%)
θ	Trade elasticity	1.5	Caldara et al. (2020)
α_p	Calvo parameter	0.75	Price adj. of one year
σ	Degree of risk aversion	2	Born and Pfeifer (2014)
ω_w^{-1}	Frisch elasticity of labor supply	1	Born and Pfeifer (2014)
σ_X	Wealth effect	0.001	Born and Pfeifer (2014)
$\bar{\ell}$	Steady state labor	0.33	Standard calibration
κ	Investment adjustment cost	6	Standard calibration
ρ_π	Taylor rule parameter wrt inflation	1.5	Standard calibration
ρ_r	Interest rate inertia	0.8	Standard calibration
g/y	Share of government expenditure in output	0.18	Bureau of Economic Analysis
Region-specific Parameters			
$\bar{\omega}$	Region-1 export intensity	0.84	Highest export intensity (0.68)
ω_1	Region-1 degree of trade openness	0.14	Import share (0.13)
ω_2	Region-2 degree of trade openness	0.08	Import share (0.08)
$\hat{\omega}_1$	Region-1 home bias	0.62	model's steady state
$\hat{\omega}_2$	Region-2 home bias	0.62	model's steady state
ω^*	Foreign degree of trade openness	0.25	Foreign import share (0.10)
η^*	Size of Foreign	0.5	U.S. population share (0.5)
η_1	Size of Region-1	0.315	Population share (0.63)
η_2	Size of Region-2	0.185	Population share (0.37)
$\bar{\tau}$	U.S. import tax	0.05	Foreign Trade Database

while the zero trade balance between Home and Foreign and at the regional level implies

$$\rho_{Hj-,t}^* Y_{Hj-,t}^* + \rho_{Hj,t}^* Y_{Hj,t}^* = \frac{1}{Q_{j-,t}} \rho_{F,t}^{j-} Y_{F,t}^{j-} + \frac{1}{Q_{j,t}} \rho_{F,t}^j Y_{F,t}^j, \quad (41)$$

$$\rho_{Hj,t} Q_{j-,t} Y_{Hj,t}^{j-} = \rho_{Hj-,t} Q_{j,t} Y_{Hj-,t}^j. \quad (42)$$

We assume that import tariffs follow an AR(1) process with stochastic volatility such that

$$(1 + \tau_t) = (1 - \rho_\tau) (1 + \bar{\tau}) + \rho_\tau (1 + \tau_{t-1}) + \exp(\sigma_t^\tau) \varepsilon_t^\tau, \quad (43)$$

$$\sigma_t^\tau = (1 - \rho_\sigma) \bar{\sigma}^\tau + \rho_\sigma \sigma_{t-1} + \gamma \varepsilon_t^\sigma, \quad (44)$$

where ρ_τ and ρ_σ are persistence parameters of level and volatility tariff shocks, respectively, γ drives the magnitude of tariff uncertainty shocks and $\bar{\sigma}^\tau$ is the steady-state standard deviation. The three other level shocks, namely the supply-driven TFP and the demand-driven public spending and monetary policy shocks follow an AR(1) process

$$x_t = (1 - \rho_x) \bar{x} + \rho_x x_{t-1} + \exp(\sigma^x) \varepsilon_t^x, \quad (45)$$

where ρ_x and σ^x are the persistence and standard-deviation parameters with $x = \{a, g, m\}$. All innovations $\varepsilon_t^z \sim \mathcal{N}(0, 1)$ for $z = \{\tau, \sigma, a, g, m\}$. Notice that although we are primarily interested in the effect of the uncertainty shock, the quantitative fit of our model requires to incorporate all standard sources of fluctuations, i.e. demand and supply shocks.

4. Parametrization

We now calibrate the set of parameters based on standard values in the literature, and some empirical targets. In our regional model, a period is a quarter. We assume that the two regions of the Home country as well as the Foreign country are perfectly symmetric except for their size and trade parameters, as we explain below. Table 4 reports our parametrization. Only the parameters specific to a given region or country are subscript by an index.

Common parameters We begin with parameters that are common to all locations. As standard in the literature, the discount factor β is set to 0.99 which fits an annual interest rate of 4%. The capital share α is equal to 0.33 so as to fit a labor income share of 0.67 and capital depreciation rate $\delta = 0.025$ implying an annual depreciation rate of 10%. We set the elasticity of substitution between intermediate goods, ε_p , to 11, implying a markup of 10%. The trade elasticity equals $\theta = 1.5$, which is consistent with the literature (see Caldara et al. (2020) for instance). The Calvo parameter is set to $\alpha_p = 0.75$ which corresponds to a price frequency change of one year. Household preference parameters are standard in the literature. We set the degree of risk aversion $\sigma = 2$ and calibrate the strength of the wealth effect ($\sigma_X = 0.001$) following Born and Pfeifer (2014) as for the Frisch elasticity ($\omega_w = 1$). The scale parameter ψ is set to ensure that the steady-state hours worked corresponded to $\bar{\ell} = 0.33$ of total hours. The investment adjustment cost is set to $\kappa = 6$. When it comes into the Taylor rule, we choose a standard value by assuming that the inflation

Table 5
Parametrization.

Shocks	Parameters	
ρ_τ	Tariff level shock: persistence	0.99
$\bar{\sigma}_\tau$	Tariff level shock: s.d. $\exp(\bar{\sigma}_\tau)$	$\exp(-5.83)$
ρ_σ	Tariff volatility shock: persistence	0.99
η	Tariff volatility shock: magnitude	0.28
ρ_a	TFP shock: persistence	0.76
σ_a	TFP level shock: s.d	$\log(0.008)$
ρ_g	Public spending shock: persistence	0.90
σ_g	Public spending shock: s.d	$\log(0.014)$
ρ_m	Monetary policy shock: persistence	0.93
σ_m	Monetary policy shock: s.d	$\log(0.0085)$

Note: All estimates are obtained by estimating Equation (45) from aggregated U.S. data.

reaction parameter for nominal interest rate is $\rho_\pi = 1.5$ and the interest rate inertia parameter is $\rho_r = 0.8$. We set the share of public spending on GDP in our sample mean to obtain $\bar{g}/\bar{y} = 0.18$. Finally, we set the steady-state value of the import tax on $\bar{\tau} = 5\%$ in both Home regions, which corresponds to our sample mean. We abstract from any retaliatory taxation.

Region-specific parameters We now turn to the calibration of region-specific parameters, particularly those that determine the trade openness of the two Home regions as well as the Foreign country ($\omega_1, \omega_2, \hat{\omega}_1, \hat{\omega}_2, \omega^*, \hat{\omega}$). It is worth noting that both regions face a common national import tariff, τ_t , and only differ in their exposure to international trade, captured through these parameters. This study is not the first to parametrize a two-region model. For instance, Nakamura and Steinsson (2014) and Chodorow-Reich (2020) assume that both regions are symmetric but that the region of interest corresponds to a representative average U.S. state, whereas the remaining region corresponds to the rest of the country. We follow an alternative path by parametrizing the representative region (Region 1) as a set of typical regions exposed to international trade, while Region 2 can be interpreted as the rest of the country. Therefore, contrary to these authors, we seek to exploit the heterogeneity across regions, and notably, the degree of exposure to trade. Specifically, we select the values of $\omega_1, \omega_2, \omega^*$ and $\hat{\omega}$ to fit four targets selected from the empirical analysis in Section 2, whereas $\hat{\omega}_1$ and $\hat{\omega}_2$ are deduced from the steady state of the model. We split the sample of 50 U.S. states into two groups of a same size and we assume that the representative exposed region is the one which has a degree of export intensity in 2008 above the median.¹⁰ The degree of the export intensity provides a target for calibrating $\hat{\omega}$. Targeting the sum of export intensity leads to $\hat{\omega} = 0.78$. We then select the values of ω_1 and ω_2 to target the 2008 average import-to-GDP share of each group, that is, 0.13 and 0.08, respectively). These empirical targets lead to $\omega_1 = 0.14$ and $\omega_2 = 0.09$. The model's steady-state leads to $\hat{\omega}_1 = \hat{\omega}_2 = 0.62$. Finally, we assume that the import-to-GDP share in the Foreign country is 10%, implying that $\omega^* = 0.25$. We also target the population share of the two groups of U.S. states, based on the empirical finding that our region of interest corresponds to 63% of the total U.S. population share in 2008 (and thus 0.37% for the other region). Assuming that the Foreign country accounts for 50% of the world economy ($\eta^* = 0.5$), we obtain $\eta_1 = 0.315$ and $\eta_2 = 0.185$.

Shocks parameters Finally, we parametrize the exogenous driving process outside the model using available aggregate U.S. series and estimate an AR(1) process over the sample 1980q1–2020q4. Results are reported in Table 5. First, we build a quarterly series of the U.S. tariff rate as in Caldara et al. (2020), where $\tau_t = CD_t/(M_t - CD_t)$ and CD_t represents the taxes on production and imports received from the Federal government and M_t is the real value of the total imports.¹¹ We then estimate a stochastic volatility process as shown in Eqs. (43)–(44) using the algorithm presented by Born and Pfeifer (2014) and select the mean values of the posterior distributions of the parameters. This parametrization strategy produces $\rho_\tau = 0.99$ and $\rho_\sigma = 0.99$ as well as $\gamma = 0.29$ and $\bar{\sigma}^\tau = -5.90$, which is in line with the results of Caldara et al. (2020). When it comes to the level shocks, i.e. TFP, public spending, and monetary policy shocks, we estimate the AR(1) process as in Eq. (45). The TFP series is borrowed from Fernald (2012) and corresponds to the utilization-adjusted measure. The public spending series is the real government consumption expenditures and gross investment provided from the U.S. Bureau of Economic Analysis. Both series are expressed in log and HP-filtered. We obtain $\rho_a = 0.76$ and $\sigma_a = \log(0.008)$ for TFP shocks as well as $\rho_g = 0.90$ and $\sigma_g = \log(0.014)$ for public spending shocks. A monetary policy shock is recovered by estimating a typical Taylor rule for Federal Funds Effective Rate (provided by the Board of Governors of the Federal Reserve System), the growth rate of the implicit price deflator and the GDP growth (both provided by the U.S. Bureau of Economic Analysis). We then estimate an AR(1) on the residuals and obtain $\rho_m = 0.93$ and $\sigma_m = \log(0.0085)$. Finally, we assume that the Foreign country is not directly hit by any shock.

¹⁰ The degree of the export intensity in state s is denoted by D_s , computed as $D_s = \frac{X_{s,2008}}{\sum_{i=1}^S X_{i,2008}}$, where $X_{s,2008}$ is the value of real exports per capita in state s in 2008 and corresponds to the beginning of the sample. We compute the median value in 2008 for all states and partition states with a value above the median (high intensity) or below the median (low-intensity). The degree of export intensity is 1/50 for each state, if it was distributed uniformly. States which feature a degree of export intensity larger than the median are “AK”, “CA”, “CT”, “DE”, “IL”, “IN”, “IA”, “KS”, “KY”, “LA”, “MA”, “MI”, “MN”, “NJ”, “NY”, “ND”, “OH”, “OR”, “SC”, “TN”, “TX”, “UT”, “VT”, “WA”, “WI”.

¹¹ Because our theoretical model is not a multi-sector model, it does not aim to capture sectoral heterogeneity in tariffs, as highlighted in Section 2. Therefore, we calibrate the tariff shock levels and volatility shocks using U.S. aggregate data.

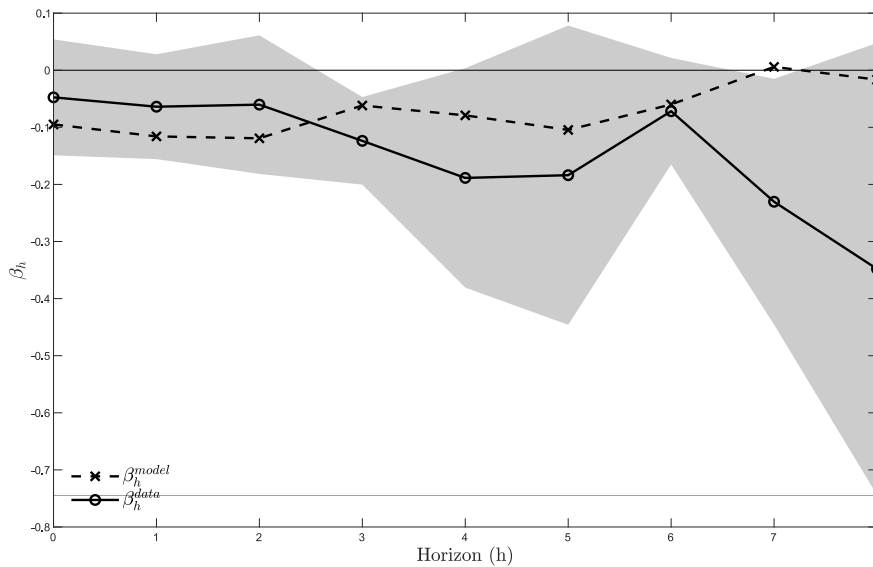


Fig. 4. Effects of regional TPU shocks on output: Data versus model.

Note: Each marker point corresponds to point estimates β_h estimated from Eq. (6) including region fixed effects for the data (solid line) and from Eq. (46) for the model (dashed line). The grey area displays the [10th, 90th] confidence interval of the empirical estimates.

5. Transmission channels of regional TPU

This section investigates the regional effects of trade policy uncertainty shocks based on the theoretical model developed in Section 3. First, we solve and simulate a theoretical model to check its goodness-of-fit. Subsequently, we analyze the Impulse Response Functions (IRFs) to trade policy uncertainty shocks, and perform a set of counterfactual exercises.

5.1. Solution methods and simulation

We solve the model by using third-order pruned perturbation methods in order to track the effects of tariff volatility shocks on the economy. All simulations are done with Dynare, see [Adjemian et al. \(2011\)](#). We first check the ability of our theoretical model to replicate empirical evidence emphasized in Section 2.3. As explained in Section 4, we parametrize the regional model to reproduce the main empirical moments related to international trade exposure which were extracted from our empirical dataset. Subsequently, we simulate the theoretical model after a sufficiently long burn-in period, that allows us to begin from the stochastic steady state. We extract the simulated series of outputs $Y_{j,t}$ for $j = \{1, 2\}$, as defined in Eq. (7), which is expressed in log. The panel structure of the theoretical model allows us to estimate the following regression

$$\log(Y_{j,t+h}) - \log(Y_{j,t-1}) = \delta_{j,h} + \delta_{t,h} + \beta_h^{model} TPU_{j,t} + v_{j,t+1}, \text{ for } h \geq 0 \quad (46)$$

for each $j = \{1, 2\}$, where $TPU_{j,t}$ corresponds to simulated volatility shocks, $\delta_{j,h}$ and $\delta_{t,h}$ are the region- and time-fixed effects, respectively. Our empirical dataset comprises 50 U.S. states, whereas our theoretical model contains two average states; we re-estimate Eq. (6) by imposing region-fixed effects in the empirical regression. This simulation-based estimation strategy has the advantage of applying a similar data-generation process than that in the empirical analysis.

Fig. 4 reports the results. Each marker corresponds to the point estimates from the data (β_h^{data}) and the model (β_h^{model}).¹² The figure shows that despite its stylized dimension, the theoretical regional model does a good job in reproducing the magnitude of the effects of tariff volatility shocks on output at several horizons. In particular, the model generates an impact elasticity of $\beta_0^{model} = -0.09$ while it is $\beta_0^{data} = -0.06$ in the data. It fails to replicate the stronger magnitude of the effects over longer horizons, but remains inside the confidence intervals over the first quarters following the shock. In the following subsection, we investigate the transmission channel for tariff volatility shocks.

5.2. IRFs analysis

We now investigate the transmission channels of a pure TPU shock in the regional model by computing the IRFs for a national volatility tariff shock and conducting a set of counterfactual exercises. We compute the response of variables to volatility shocks

¹² Therefore, the data-based markers for the horizon $h = 0, 4, 8$ correspond to the counterpart values reported in Table 1 when we include the region-fixed effect instead of the state-fixed effect. Footnote 19 defines the region dummy variable.

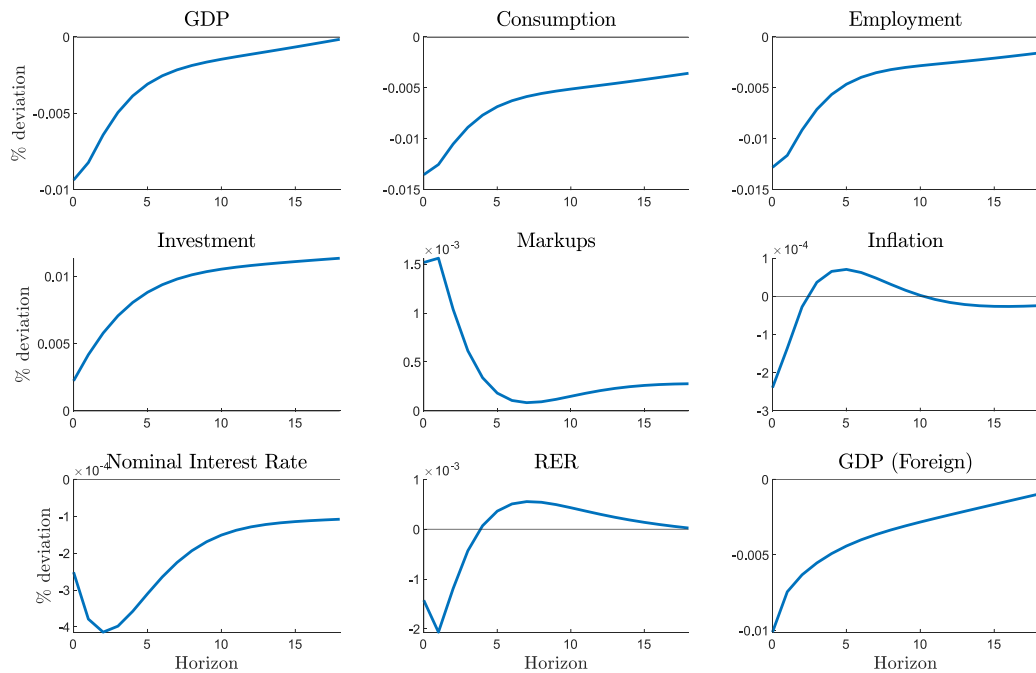


Fig. 5. IRFs to a TPU shock: Baseline. Note: IRFs are computed in deviation from the ergodic mean and multiplied by 100.

deviating from the EMAS, with all the other shocks shut down.¹³ Henceforth, we display the responses in the representative region, i.e. the most exposed region to international trade and we investigate the main drivers of these effects.¹⁴

5.2.1. Baseline calibration

Fig. 5 shows the IRFs in the representative exposed region to a national TPU shock that hits the Home economy. We find that an increase in import tariff volatility generates a drop in consumption and a recession, while there is a surge in investment. In Keynesian models in which production is determined by demand, a decrease in consumption leads to a fall in production. The investment reaction depends on the relative magnitude of these two decreases. Here, the fall in consumption is so large relative to the realized output that the unconsumed output available for investment increases. This results from the high persistence of the uncertainty shock. For a lower value of persistence, we find the usual co-movement between consumption, investment, and production in response to uncertainty shocks without affecting our results. Higher uncertainty about tariffs in the Home economy leads to a collapse of its imports, meaning for the Foreign economy, a contraction in demand for its own production. Consequently, the Foreign country also enters a recession, generating a feedback effect through a reduction in Home exports. Overall, a reduction in the aggregate demand generates an increase in markups and a drop in inflation. Unlike Caldara et al. (2020), our model does not incorporate firms' export participation decisions. Even without this mechanism, our model captures the main expected effects of uncertainty shock to tariffs, as the authors showed that the extensive margin of exports primarily affects the magnitude of the responses rather than the transmission channels.¹⁵

5.2.2. Precautionary behavior

It is well documented that the recessionary effects of uncertainty shocks are explained by the precautionary behavior of agents, particularly the precautionary saving motive and precautionary pricing behavior of firms (see Bianchi et al. (2023) for

¹³ Technically, we simulate the model starting from a deterministic steady state, with all shocks set to zero and burn in the 10 000 first to obtain an ergodic mean without shock (EMAS). We simulate the model once again, impulsing the volatility innovation ε_t^σ after a burn-in period. Then the IRFs were expressed in deviation from EMAS.

¹⁴ Although the model is parametrized to capture heterogeneity in trade exposure across the two regions, their responses follow a similar pattern and magnitude. House et al. (2021) show that interstate trade generates stronger reaction of the economic activity to shocks. We complement their results by finding that it reduces heterogeneity, because the effects of shocks propagate among regions through trade. In the online appendix, we report the responses of both regions.

¹⁵ In the online appendix, we report the IRFs when: (i) we adopt King et al. (1988)'s preferences by setting $\sigma_x = 1$; (ii) the monetary policy is more aggressive, i.e. $\rho_\pi = 5$; (iii) prices are more flexible ($\alpha_p = 0.4$). We find that while the magnitude of the effects vary, the typical transmission channels of uncertainty shocks holds in all specifications. We also consider a worldwide uncertainty shock on trade policy in the model, i.e. both Home and Foreign countries impose import tariffs and they are hit by an exogenous volatility shock in order to capture the effects on the export side. We find that a global uncertainty shock mostly raises the magnitude of the recession in both Home regions while the difference between the regional IRFs remains similar.

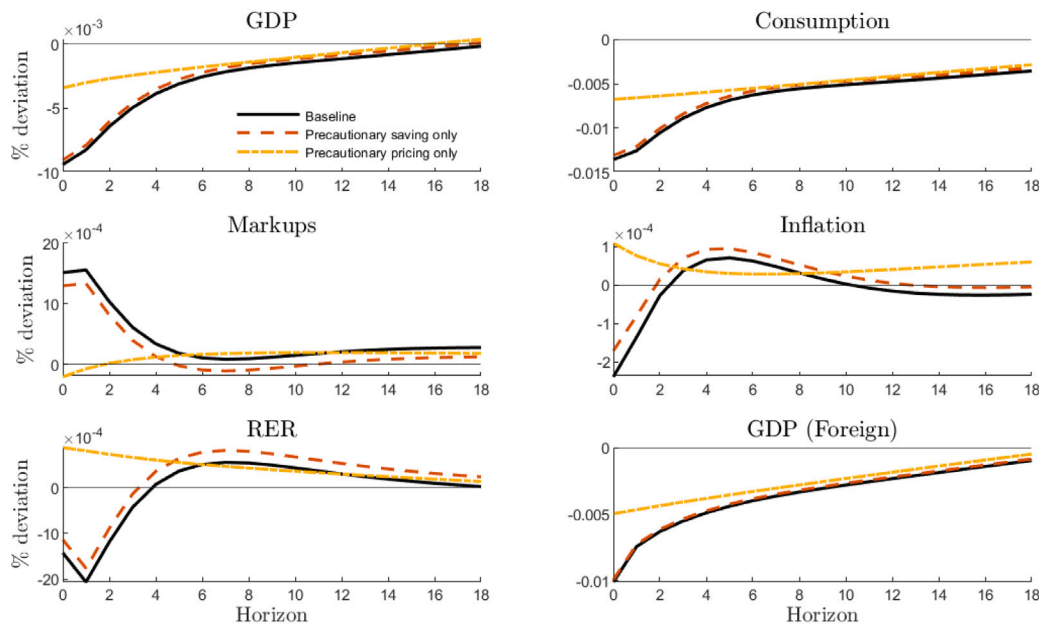


Fig. 6. IRFs to a TPU shock: Precautionary motives.

Note: IRFs are computed in deviation from the ergodic mean and multiplied by 100. The solid lines correspond to our baseline setup. The dashed orange lines corresponds to a model with linearized pricing Eq. (14). The dotted-dashed yellow lines display the case with constant stochastic discount factor in Eq. (47).

details). First, the convex marginal utility of consumption implies that households who face uncertainty about the future economic prospective postpone their current consumption and increase their savings to smooth consumption over time. As a consequence of this precautionary saving channel, a higher volatility of import tariffs generates second-order effects, leading to a reduction in private consumption. The rise in labor supply arising from “precautionary labor supply” Basu and Bundick (2017) is attenuated through our parametrization of the zero-wealth effect on labor supply (σ_X small) combined with the presence of nominal rigidities. Second, trade policy uncertainty makes firms more uncertain about their expected real marginal costs because import tariffs have a direct effect on the real marginal cost, as shown in Eq. (18), and as emphasized by Caldara et al. (2020). Therefore, the variance of the desired future price is higher, which makes firms more “prudent” in their pricing decisions by setting a higher price than what they would normally do in the absence of uncertainty. This precautionary pricing behavior leads firms to increase markups, which reduces output because firms are demand-constrained.

Our model allows us to disentangle the contributions of these two second-order moment channels in explaining recessions. Fig. 6 displays the IRFs of output, consumption, markups, and inflation in the baseline model (solid lines), when we shut down the precautionary pricing channel (by linearizing the New-Keynesian Phillips Curve (14), dashed orange lines), and the precautionary savings channel (by imposing a constant stochastic discount factor in the Euler Eq. (47) and dotted-dashed yellow lines) in this region.¹⁶

Fig. 6 shows that precautionary pricing behavior explains a very small portion of the recession, while most of the magnitude of output and consumption is explained by the presence of the precautionary saving motive. When the precautionary pricing behavior only is active, we find that markups increase (after one period), but by a small amount, accompanied with an increase in inflation and the size of the recession is twice smaller. The reduction in inflation observed in the baseline is mostly explained by the negative effect of precautionary saving on aggregate demand that generates a real exchange rate appreciation, the later being also explained by the drop in Foreign activity that results from a reduction in domestic exports. In total, the fall in output and consumption is close to the baseline when only the precautionary saving motive is present.

5.2.3. Assessing the role of trade exposure

Our previous analysis allowed us to emphasize the transmission channels of uncertainty shocks that rely on the closed-economy structure of the model –and, more particularly, the precautionary saving motive. However, the regional effects of TPU shocks depend on the regional degree of trade exposure. We assess the open-economy structure of the model in more detail by plotting the IRFs of a set of variables to a national TPU shock when we alternately modify parameters ω_j and ω^* .

¹⁶ For the last counterfactual, we assume a linear utility function for households. In addition to ignoring the precautionary saving motive, we assume that households are no longer risk averse. In order to focus on precautionary saving behaviors, we impose a constant stochastic discount factor in the Euler equation only, keeping it endogenous and time varying in other equations.

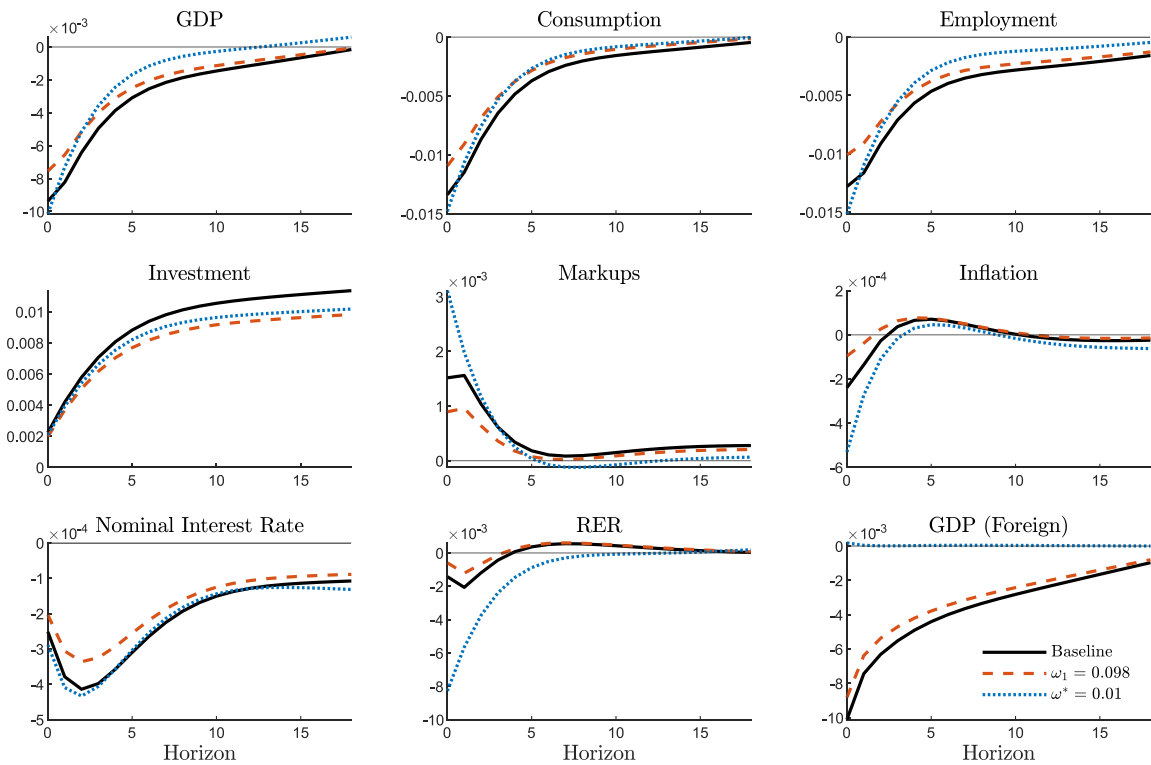


Fig. 7. IRFs to a TPU shock: International trade channels.

Note: IRFs are computed in deviation from the ergodic mean and multiplied by 100. The solid line displays the response for the baseline calibration. The dashed orange line shows the IRFs for a lower import share ($\omega_1 = 0.098$). The dotted blue line represents the IRFs when we shut down the feedback effect from the foreign country ($\omega^* = 0.01$).

The solid line in Fig. 7 represents the responses for the baseline calibration. The dashed orange line represents the IRFs with a lower import share; i.e. ω_1 is set to the alternative region's value. In that case, production technology in the region is less dependent on imported goods, which, in turn, results in a demand function for imported goods that is less sensitive to tariff volatility. Therefore, the transmission channels of uncertainty shocks are attenuated through dampened appreciation of the real exchange rate and this generates a slightly more limited recession. Interestingly, we also find using our empirical dataset that states which have an import share larger than the median suffer from a stronger recession, although this effects is hardly significant (see online appendix). The dotted blue lines in Fig. 7 indicates the IRFs when we assume that the Foreign country has a large home bias (ω^* is small). This allows us to assess the role of the feedback effect since we shut down the trade channel between the Foreign country and the region. Because the Foreign country consumes domestic goods only, Foreign output becomes insensitive to Home uncertainty shocks. This corresponds to the non-feedback case. The Home country experiences still a recession coming from the precautionary behavior of agents, as explained above. Looking at the dynamics of output, consumption, and employment, the impact responses are the same in the no-feedback and the feedback case. Importantly, the feedback effect makes the dynamics responses of these variables more persistent, because the drop in Foreign output generates a long-lasting reduction in the demand for Home goods emanating from Foreign firms. This leads us to conclude that trade connections between Home and Foreign raise duration of the recessionary effects of TPU shocks on the economy.

6. Conclusion

This study examines the effects of trade policy uncertainty shocks, both empirically and theoretically, using a regional approach. We build a measure of U.S. states' exposure to trade policy uncertainty, and estimate the effects of a related shock on GDP and employment. We show that a positive shock to regional TPU exposure is recessionary. We rationalize this result using a two-region open-economy model. We find that the precautionary behavior of agents, and more precisely, the precautionary saving motive, acts as the main driver of the recessionary effect of TPU shock; this effect is reinforced by a high exposure to tariffs. In contrast, the feedback effect resulting from trade connections with a foreign country mostly affects the persistence of responses to TPU shocks. Our findings complement previously obtained results at different aggregation levels, as for instance at the country or the firm level. This analysis can be pursued in the analysis of the local policies that mitigate the effects of tariffs at the national level.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Households first-order conditions

Maximizing the discounted lifetime utility measure (33) subject to the definition (34), the budget constraint (35), the law of capital accumulation (36) yields the following first-order conditions

$$1 = \beta E_t \left\{ \frac{\lambda_{j,t+1}}{\lambda_{j,t}} \frac{R_{j,t}}{\pi_{j,t+1}} \right\}, \quad (47)$$

$$(c_{j,t} - \psi \mathcal{X}_{j,t} (\ell_{j,t})^{\omega_w})^{-\sigma} + \sigma_X v_{j,t} c_{j,t}^{\sigma_X - 1} \mathcal{X}_{j,t}^{1 - \sigma_X} = \lambda_{j,t}, \quad (48)$$

$$v_{j,t} + \psi (\ell_{j,t})^{\omega_w} (c_{j,t} - \psi \mathcal{X}_{j,t} (\ell_{j,t})^{\omega_w})^{-\sigma} = \beta (1 - \sigma_X) E_t \left\{ v_{j,t+1} c_{j,t+1}^{\sigma_X} \mathcal{X}_{j,t}^{-\sigma_X} \right\}, \quad (49)$$

$$\psi \omega_w (\ell_{j,t})^{\omega_w - 1} \mathcal{X}_{j,t} (c_{j,t} - \psi \mathcal{X}_{j,t} (\ell_{j,t})^{\omega_w})^{-\sigma} = \lambda_{j,t} w_{j,t}, \quad (50)$$

$$\varsigma_{j,t} = \left[1 - \Phi \left(\frac{i_{j,t}}{i_{j,t-1}} \right) - \Phi' \left(\frac{i_{j,t}}{i_{j,t-1}} \right) \frac{i_{j,t}}{i_{j,t-1}} + \beta E_t \left\{ \frac{\lambda_{j,t+1} \varsigma_{j,t+1}}{\lambda_{j,t} \varsigma_{j,t}} \left(\frac{i_{j,t+1}}{i_{j,t}} \right)^2 \Phi' \left(\frac{i_{j,t+1}}{i_{j,t}} \right) \right\} \right]^{-1}, \quad (51)$$

$$1 = \beta E_t \left\{ \frac{\lambda_{j,t+1}}{\lambda_{j,t}} \left[\frac{z_{j,t+1} + (1 - \delta) \varsigma_{j,t+1}}{\varsigma_{j,t}} \right] \right\}, \quad (52)$$

with $\varsigma_{j,t} \equiv \check{z}_{j,t} / \lambda_{j,t}$, the relative price of capital and with $\Phi \left(\frac{i_{j,t}}{i_{j,t-1}} \right) = \frac{\kappa}{2} \left(\frac{i_{j,t}}{i_{j,t-1}} - 1 \right)^2$ and $\Phi' \left(\frac{i_{j,t}}{i_{j,t-1}} \right) = \kappa \frac{i_{j,t}}{i_{j,t-1}} \left(\frac{i_{j,t}}{i_{j,t-1}} - 1 \right)$, and where λ_t , v_t and $\check{z}_t$ denote the Lagrangian multipliers associated to the budget constraint (35), the definition (34), and the law of motion of capital (36), respectively.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jinteco.2025.104134>.

Data availability

Replication file (Original data) (Mendeley Data)

References

- Adão, R., Kolesár, M., Morales, E., 2019. Shift-share designs: Theory and inference. *Q. J. Econ.* 134 (4), 1949–2010.
- Adjemian, S., Bastani, H., Juillard, M., Mihoubi, F., Perendia, G., Ratto, M., Villemot, S., 2011. Dynare: Reference Manual, Version 4. Dynare working papers 1, CEPREMAP.
- Ahir, H., Bloom, N., Furceri, D., 2022. The world uncertainty index. NBER Working Paper 29763.
- Alessandria, G.A., Khan, S.Y., Khederlarian, A., Ruhl, K.J., Steinberg, J.B., 2024. Trade war and peace: U.S.-China trade and tariff risk from 2015–2050. NBER Working Paper 32150.
- Auerbach, A.-J., Gorodnichenko, Y., Murphy, D., 2020. Local fiscal multipliers and fiscal spillovers in the USA. *IMF Econ. Rev.* 68, 195–229.
- Autor, D., Dorn, D., Hanson, G.H., 2013. The China syndrome: Local labor market effects of import competition in the United States. *Am. Econ. Rev.* 103 (6), 2121–2168.
- Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. *Q. J. Econ.* 131 (4), 1593–1636.
- Baker, S.R., Davis, S.J., Levy, J.A., 2022. State-level economic policy uncertainty. *J. Monet. Econ.* 132, 81–99.
- Basu, S., Bundick, B., 2017. Uncertainty shocks in a model of effective demand. *Econometrica* 85 (3), 937–958.
- Beraja, M., Fuster, A., Hurst, E., Vavra, J., 2018. Regional heterogeneity and the refinancing channel of monetary policy. *Q. J. Econ.* 134 (1), 109–183.
- Beraja, M., Hurst, E., Ospina, J., 2019. The aggregate implications of regional business cycles. *Econometrica* 87 (6).
- Bianchi, F., Kung, H., Tirsikh, M., 2023. The origins and effects of macroeconomic uncertainty. *Quant. Econ.* 14 (3), 855–896.
- Born, B., Pfeifer, J., 2014. Policy risk and the business cycle. *J. Monet. Econ.* 68, 68–85.
- Bown, C.P., Kolb, M., 2023. Trump's trade war timeline: An up-to-date guide. pp. 1–23, PIIE Trade and Investment Policy Watch.
- Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A., Raffo, A., 2020. The economic effects of trade policy uncertainty. *J. Monet. Econ.* 109, 38–59.
- Calvo, G.A., 1983. Staggered prices in a utility-maximizing framework. *J. Monet. Econ.* 12 (3), 383–398.
- Canova, F., 2024. Should we trust cross-sectional multiplier estimates? *J. Appl. Econometrics* 39 (4), 589–606.
- Chodorow-Reich, G., 2019. Geographic cross-sectional fiscal spending multipliers: What have we learned? *Am. Econ. Journal: Econ. Policy* 11 (2), 1–34.
- Chodorow-Reich, G., 2020. Regional data in macroeconomics: Some advice for practitioners. *J. Econom. Dynam. Control* 115, 103875.
- Chodorow-Reich, G., Feiveson, L., Liscow, Z., Woolston, W.G., 2012. Does state fiscal relief during recessions increase employment? evidence from the American recovery and reinvestment acts. *Am. Econ. Journal: Econ. Policy* 4 (3), 118–145.
- Dupor, B., Guerrero, R., 2017. Local and aggregate fiscal policy multipliers. *J. Monet. Econ.* (ISSN: 0304-3932) 92, 16–30.
- Elkamhi, R., Jo, C., Salerno, M., 2024. Measuring “state-level” economic policy uncertainty. *J. Financ. Quant. Anal.* 59 (7), 3230–3266.
- Fajgelbaum, P.D., Goldberg, P.K., Kennedy, P.J., Khandelwal, A.K., 2020. The return to protectionism. *Q. J. Econ.* 135 (1), 1–55.
- Fernald, J., 2012. A quarterly, utilization-adjusted series on total factor productivity. Federal Reserve Bank of San Francisco. Working Paper Series 2012-19.

- Fernández-Villaverde, J., Guerrón-Quintana, P.A., 2020. Uncertainty shocks and business cycle research. *Rev. Econ. Dyn.* 37, 118–S146.
- Fernández-Villaverde, J., Guerrón-Quintana, P., Kuester, K., Rubio-Ramírez, J., 2015. Fiscal volatility shocks and economic activity. *Am. Econ. Rev.* 105 (11), 3352–3384.
- Fernández-Villaverde, J., Guerrón-Quintana, P., Rubio-Ramírez, J.F., Uribe, M., 2011. Risk matters: The real effects of volatility shocks. *Am. Econ. Rev.* 101 (6), 2530–2561.
- Firooz, H., Leduc, S., Liu, Z., 2025. Reshoring, automation, and labor markets under trade uncertainty. *J. Int. Econ.* (Forthcoming).
- Goldsmith-Pinkham, P., Sorkin, I., Swift, H., 2020. Bartik instruments: What, when, why, and how. *Am. Econ. Rev.* 110 (8), 2586–2624.
- Handley, K., 2014. Exporting under trade policy uncertainty: Theory and evidence. *J. Int. Econ.* 94 (1), 50–66.
- Handley, K., Limão, N., 2015. Trade and investment under policy uncertainty: Theory and firm evidence. *Am. Econ. J.: Econ. Policy* 7 (4), 189–222.
- Handley, K., Limão, N., 2017. Policy uncertainty, trade, and welfare: Theory and evidence for China and the United States. *Am. Econ. Rev.* 107 (9), 2731–2783.
- Handley, K., Limão, N., 2022. Trade policy uncertainty. *Annu. Rev. Econ.* 14, 363–395.
- Hazell, J., Herreno, J., Nakamura, E., Steinsson, J., 2022. The slope of the phillips curve: Evidence from US states. *Q. J. Econ.* 137 (3), 1299–1344.
- House, C.L., Proebsting, C., Tesar, L.L., 2018. Quantifying the benefits of labor mobility in a currency union. NBER Working Paper 25347.
- House, C.L., Proebsting, C., Tesar, L.L., 2021. Regional effects of exchange rate fluctuations. *J. Money, Credit, Bank.* 52, 429–463.
- King, R.G., Plosser, C.I., Rebelo, S.T., 1988. Production, growth and business cycles : I. The basic neoclassical model. *J. Monet. Econ.* 21 (2–3), 195–232.
- Leduc, S., Liu, Z., 2016. Uncertainty shocks are aggregate demand shocks. *J. Monet. Econ.* 82, 20–35.
- Nakamura, E., Steinsson, J., 2014. Fiscal stimulus in a monetary union: Evidence from US regions. *Am. Econ. Rev.* 104 (3), 753–792.
- Santacreu, A.M., Spasi, M., Zhang, J., 2023. What determines state heterogeneity in response to US tariff changes?. Mimeo.
- Shoag, D., Veuger, S., 2016. Uncertainty and the geography of the great recession. *J. Monet. Econ.* 84, 84–93.
- Steinberg, J.B., 2019. Brexit and the macroeconomic impact of trade policy uncertainty. *J. Int. Econ.* 117, 175–195.